



Portfolio

XU XIAOCHI 徐晓驰

TONGJI UNIVERSITY 同济大学

URBAN DESIGN 城市设计

2020—2024 2024-2020

• Education

Tongji University, Shanghai

2020.9-2025.7

Bachelor of Engineering, Urban Design

GPA: 4.68/5 91.83/100

• Academic Experience

2nd Level Scholarship, Tongji University

2021-22/2022-23

International Design Summer Camp, Vienna, Prague, Berlin

2023.8-10

TU Wien, CTU | Supervisor: Prof. Kai Xu, Prof. Gang Liu

- Designed for Arsenal New City Transformation.
- Researched in Vienna, Prague, Berlin Urban History and Morphology.

CAUP International Summer School, Shanghai

2021.5-6

Tongji University | Supervisor: Prof. Emilie Gascon, Prof. Jose Manuel Almondovar Melendo, Prof. Huang Huang.

- Developed innovative urban planning solutions that promote health and resilience in Cities, focusing on creating comprehensive designs for Yangpu Fuxing Island.
- As group leader, cooperated with students from 6 different countries, responsible for organising, idea gathering, design sketch, master plan drawing and presentation.

• Intern Experience

Urban Designer Intern: Jinqiao Real Estate Group

2023.7-8

- Participate in program discussion, project symposium and on-site research and investigation, eg. curtain wall panel Acceptance in Jinwan Wuqi Mountain project.
- Data collecting, text organizing and public tweets writing.

Freelance Intern: Landscape Space Sub-laboratory, Tongji University

2023.4-5

- Landscape Analysis and Renovation of Changwu Campus
- Site research, mapping and conceptual design; Responsible for master plan sketching using Illustrator.

• Social Experience

Outstanding Officer of Student Union

2022-2023

Student Development Department, CAUP Student Union: Head of Research and Study Planning Team.

- Organized and planned Office Tour and Citywalk.

Vice Monitor of Urban Design Class of 2020

2021-2023

- Assisting the class monitor in the annual evaluation of excellence and other class activities.

Volunteer: 2021 International Student Conference on Environment and Sustainability, UN

2021.6



Xiaochi Xu

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Professional Skills

SketchUp
●●●●●●●●●●

Rhino
●●●●●●○○○○

ArcGIS
●●●●●●●○○○

Photoshop/Illustrator/
InDesign
●●●●●●●○○○

AutoCAD
●●●●●●●●○○

Enscape/Lumion
●●●●●●○○○○

Model Making
●●●●●●●●●●

Personal Skills

Communication
●●●●●●●●●●

Team Working
●●●●●●●●●●

Mandarin
●●●●●●●●●●

English: IELTS 8
●●●●●●●●○○

German: B1
●●●●●○○○○○

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01

超级步行街区 Super Pedestrian District

城市设计专业大三下课程设计

指导老师：戴代新 孙彤宇 许 凯

小组成员：徐晓驰 阳馨怡 余映萱

Phase I：超级步行街区城市设计

Urban Design in Super Pedestrian District

设计任务：拟设置 1kmX1km 的范围进行城市设计，旨在探索一种将若干个街区作为一个集合，并在其区域内实现步行优先的区域步行化规划模式。在设计的过程中，需要探讨用地、交通、步行系统、公交系统、公共设施配置、开发强度、空间结构、功能混合可能性等问题。

设计思路说明：整体的思路是引入一条斜向的河流，并且放置在地块的偏心位置，以打造出兼具地块商业价值和休闲价值的滨水商业集聚区；其次沿河流方向设置两条斜轴，格网叠加在斜轴之上形成整体的路网结构。功能上来看，以斜轴交叉点为中心向外辐射，功能大致依次为商办、商住混合、居住，其中依据人口及区域组团情况设置公共服务功能，并形成舒适的水绿体系和交错联通的步行系统。

* 选择地块东北方向部分地块绘制控制性详细规划图则。



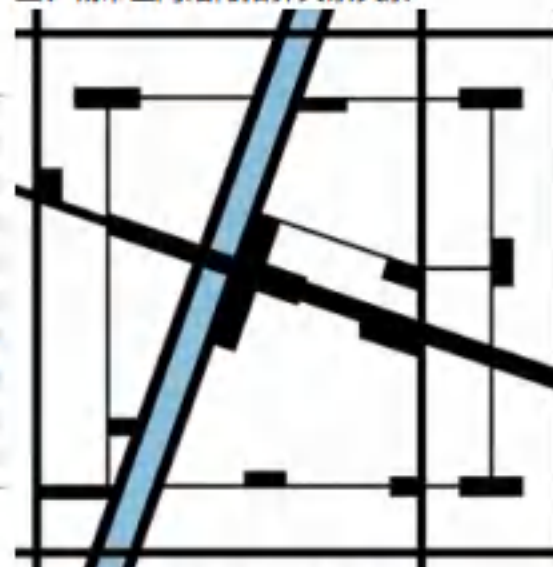
一、城市图底关系图



二、城市机动交通系统图



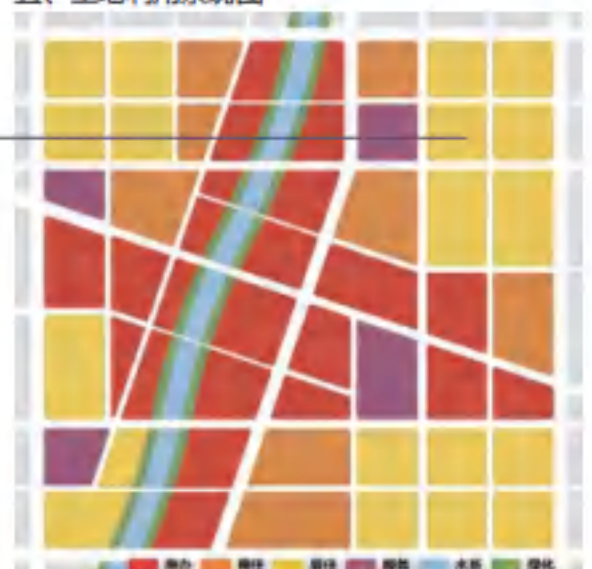
三、城市空间结构拓扑关系图



四、城市公共交通系统图



五、土地利用系统图



六、15分钟步行圈底层功能分布



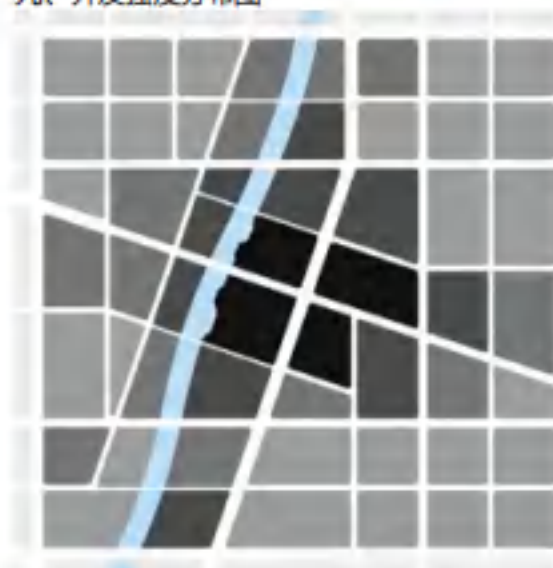
七、城市步行系统和公共空间节点分布图



八、城市绿化和水系系统图



九、开发强度分布图



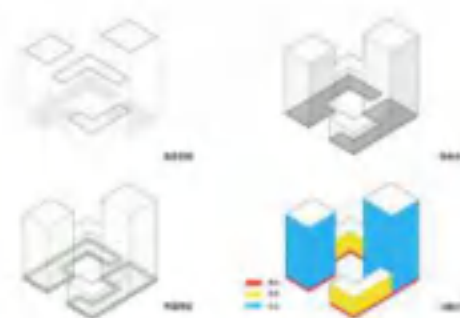
十、建筑物高度分布图



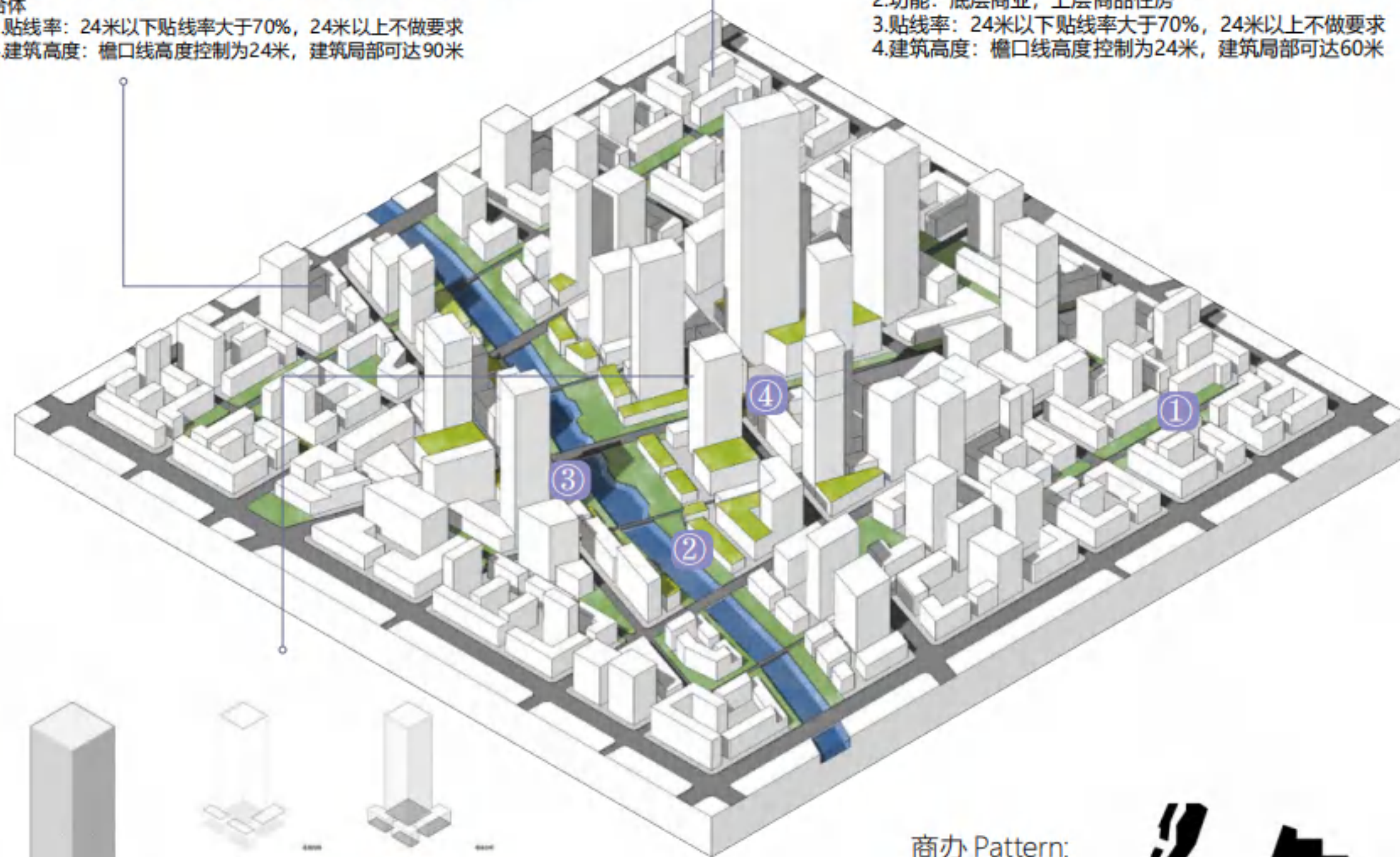
表十图分析



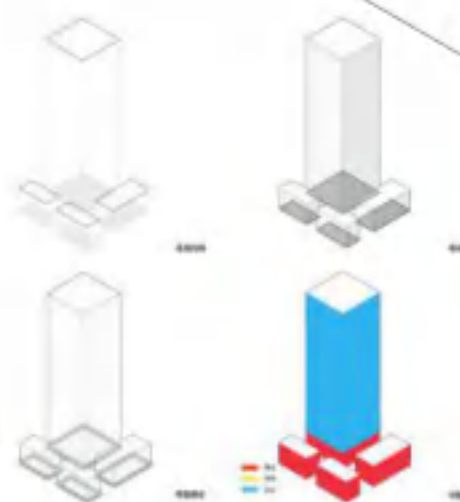
- 1.经济指标: 容积率7.0-7.5, 建筑密度不高于40%
- 2.功能: 底层商业, 以居住为主, 混合少量办公和商业的住宅综合体
- 3.贴线率: 24米以下贴线率大于70%, 24米以上不做要求
- 4.建筑高度: 檐口线高度控制为24米, 建筑局部可达90米



- 1.经济指标: 容积率4.0-4.5, 建筑密度不高于45%
- 2.功能: 底层商业, 上层商品住房
- 3.贴线率: 24米以下贴线率大于70%, 24米以上不做要求
- 4.建筑高度: 檐口线高度控制为24米, 建筑局部可达60米



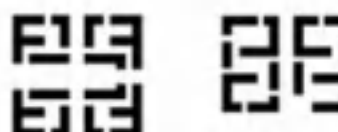
- 1.经济指标: 容积率8.0-15.0, 建筑密度不高于35%
- 2.功能: 商业办公混合综合体
- 3.贴线率: 24米以下贴线率大于60%, 24米以上不做要求
- 4.建筑高度: 檐口线高度控制为24米, 高层建筑不做控制



商办 Pattern:



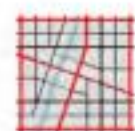
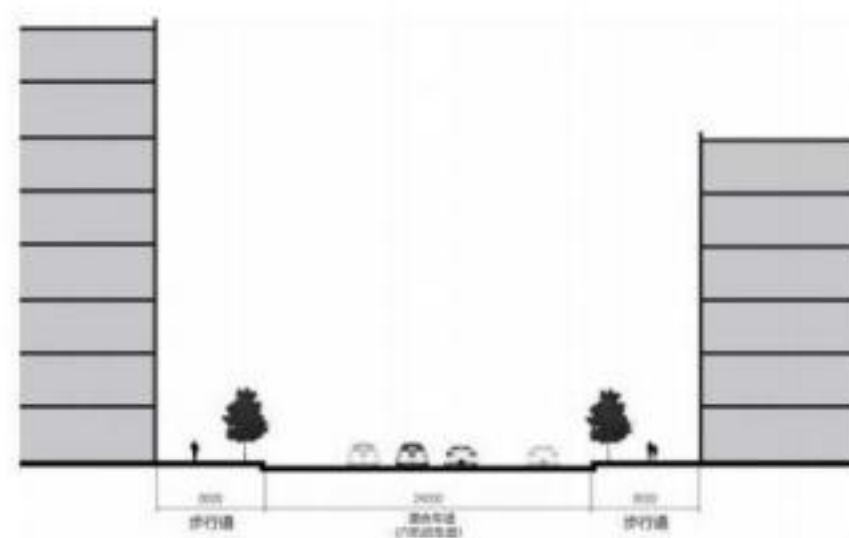
住宅 Pattern:



商住 Pattern:



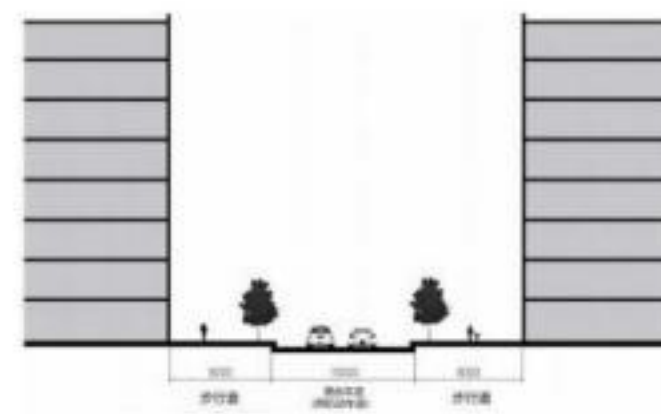
主要道路类型和断面



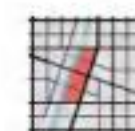
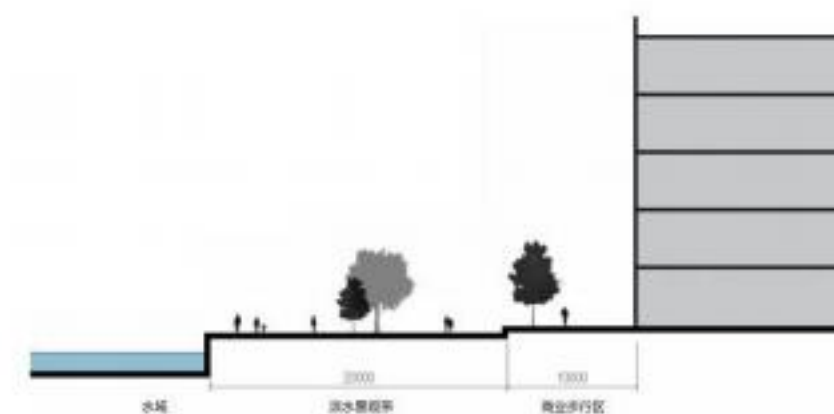
主干道街道断面



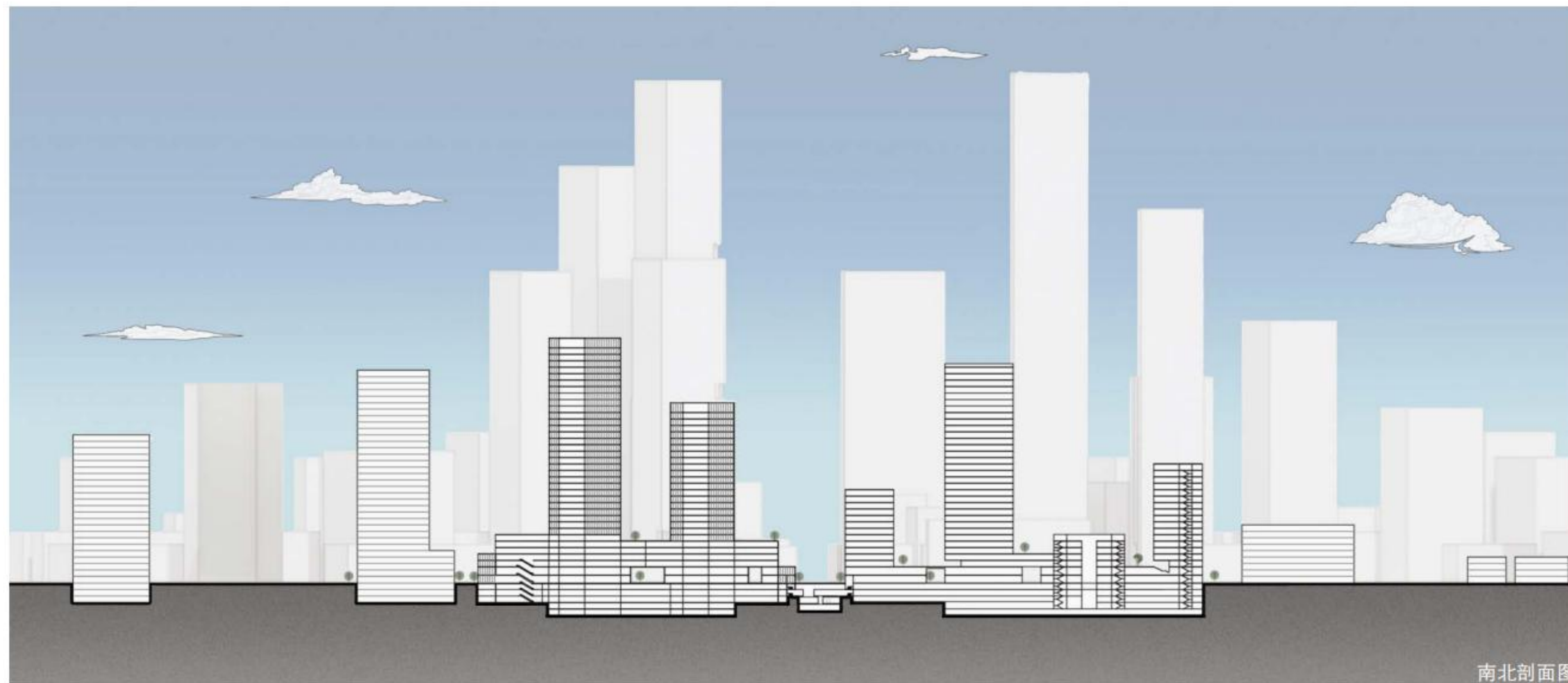
次干道街道断面



居住区支路街道断面



滨水步行区街道断面

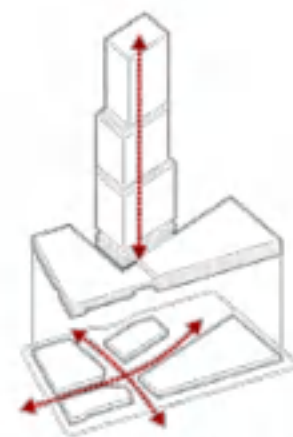


南北剖面图

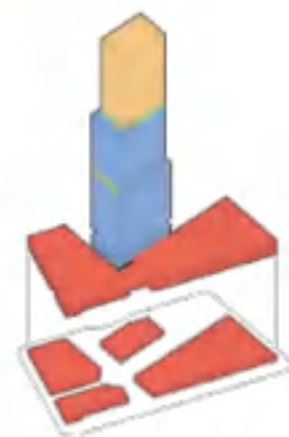
Phase II : 超级步行街区——城市综合体建筑设计

Architectural Design of Urban Complex in Super Pedestrian District

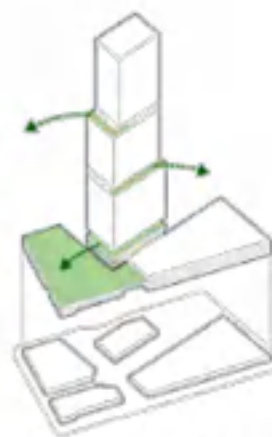
设计说明：综合体选址在中心绿轴一侧，是十字轴心的 TOD 组成之一。底层裙房为商业综合体，塔楼放置商业和办公功能，形成功能混合的商办混合街区。街区内部在地面层形成十字通廊，同时打通地下空间、开放屋顶平台，垂直方向上也为用户提供丰富的路径选择和面向滨水区的景观视线。



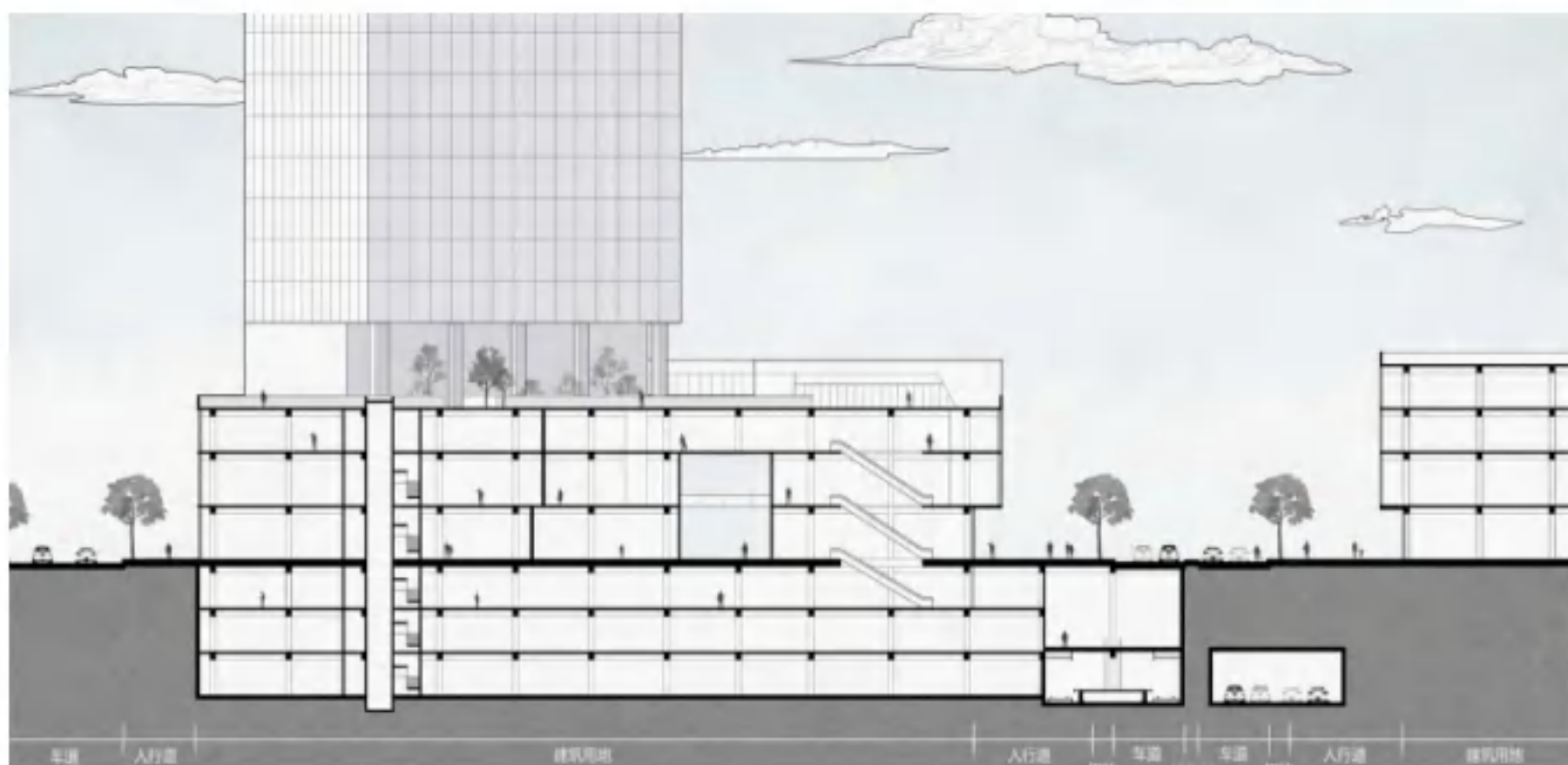
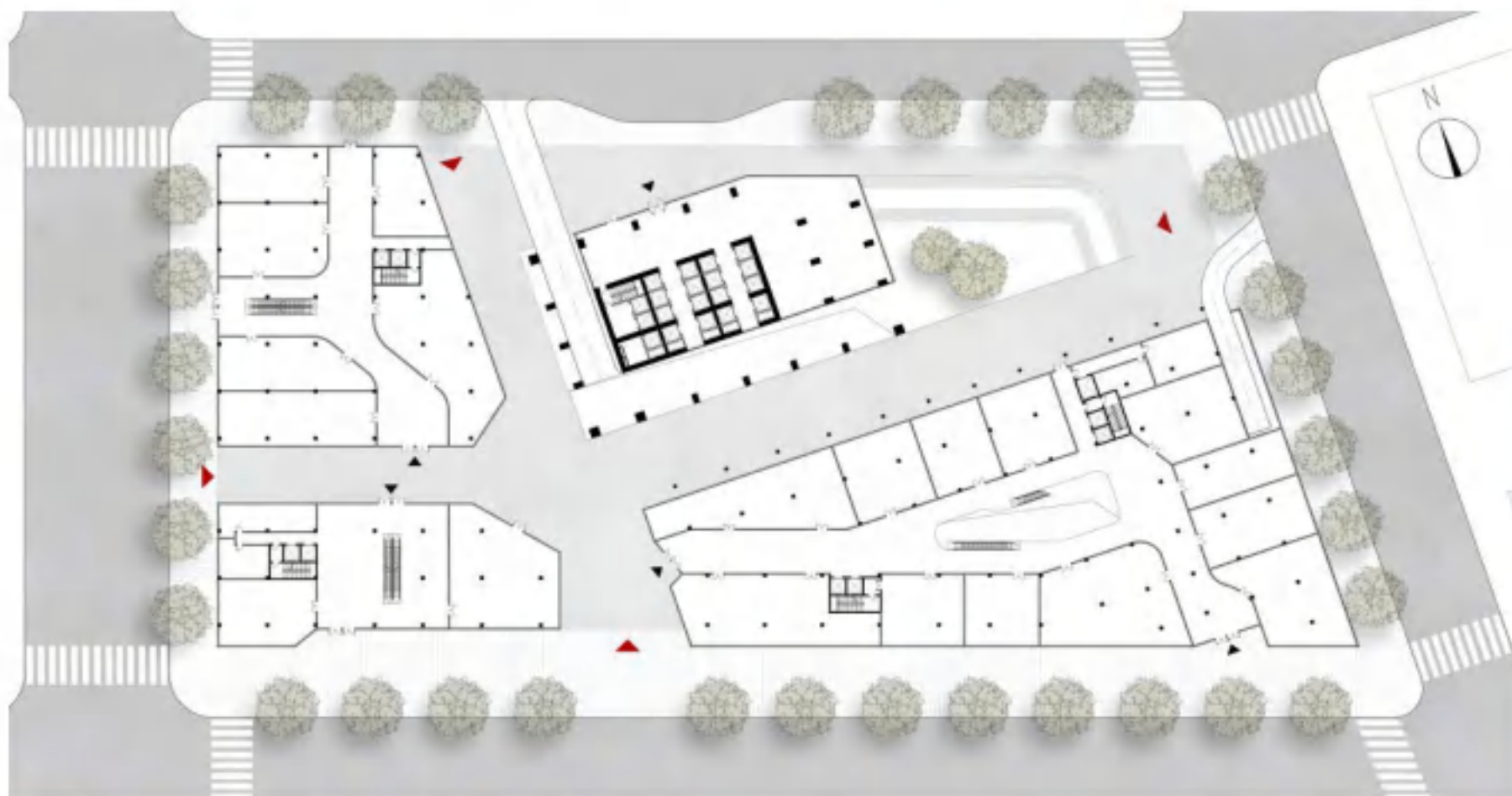
流线组织



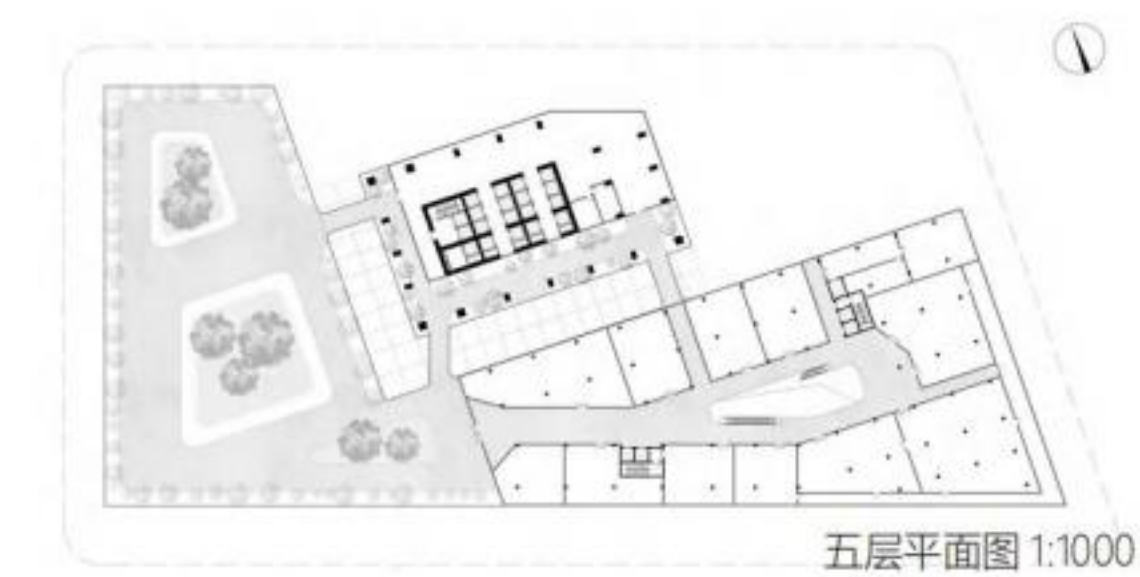
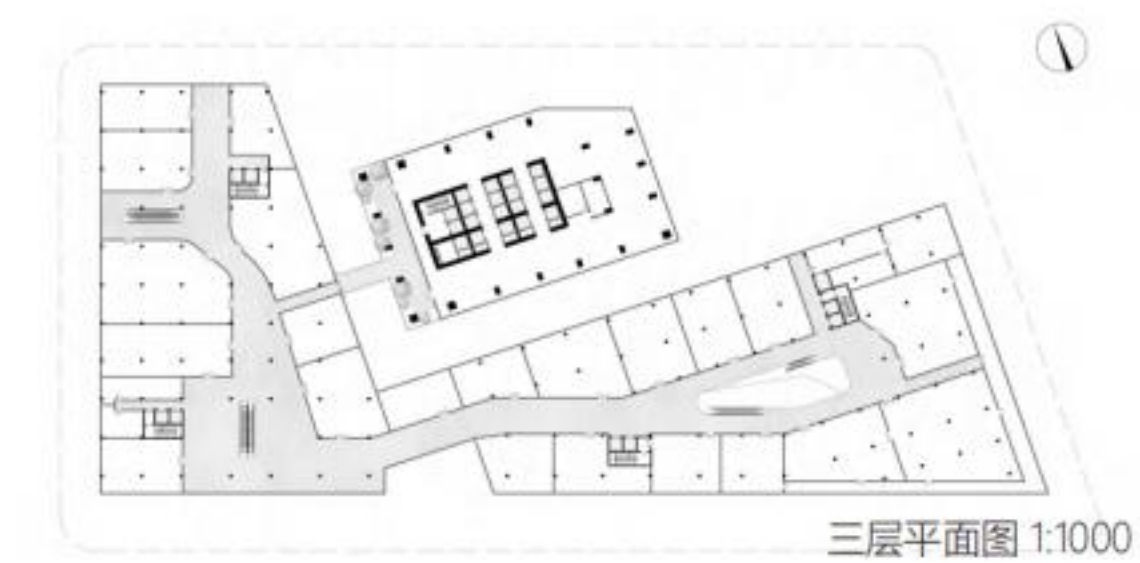
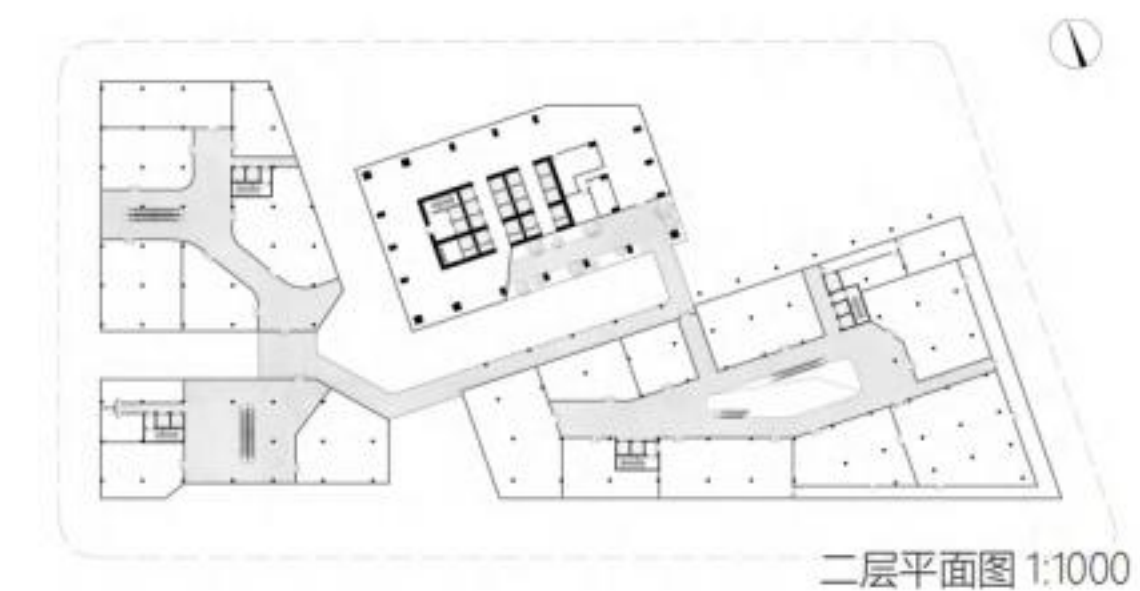
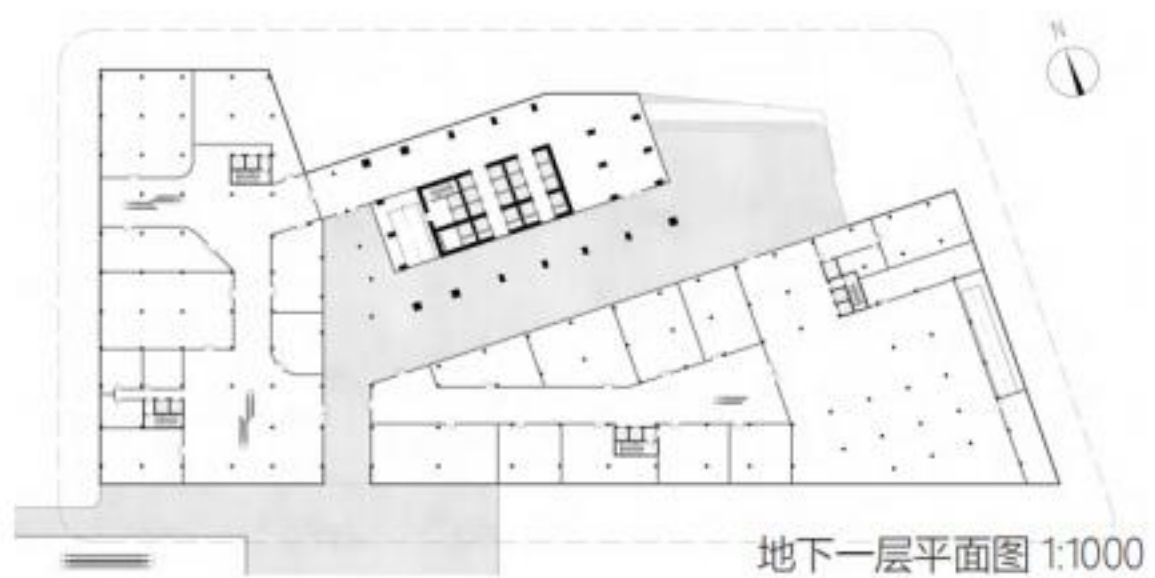
功能分布



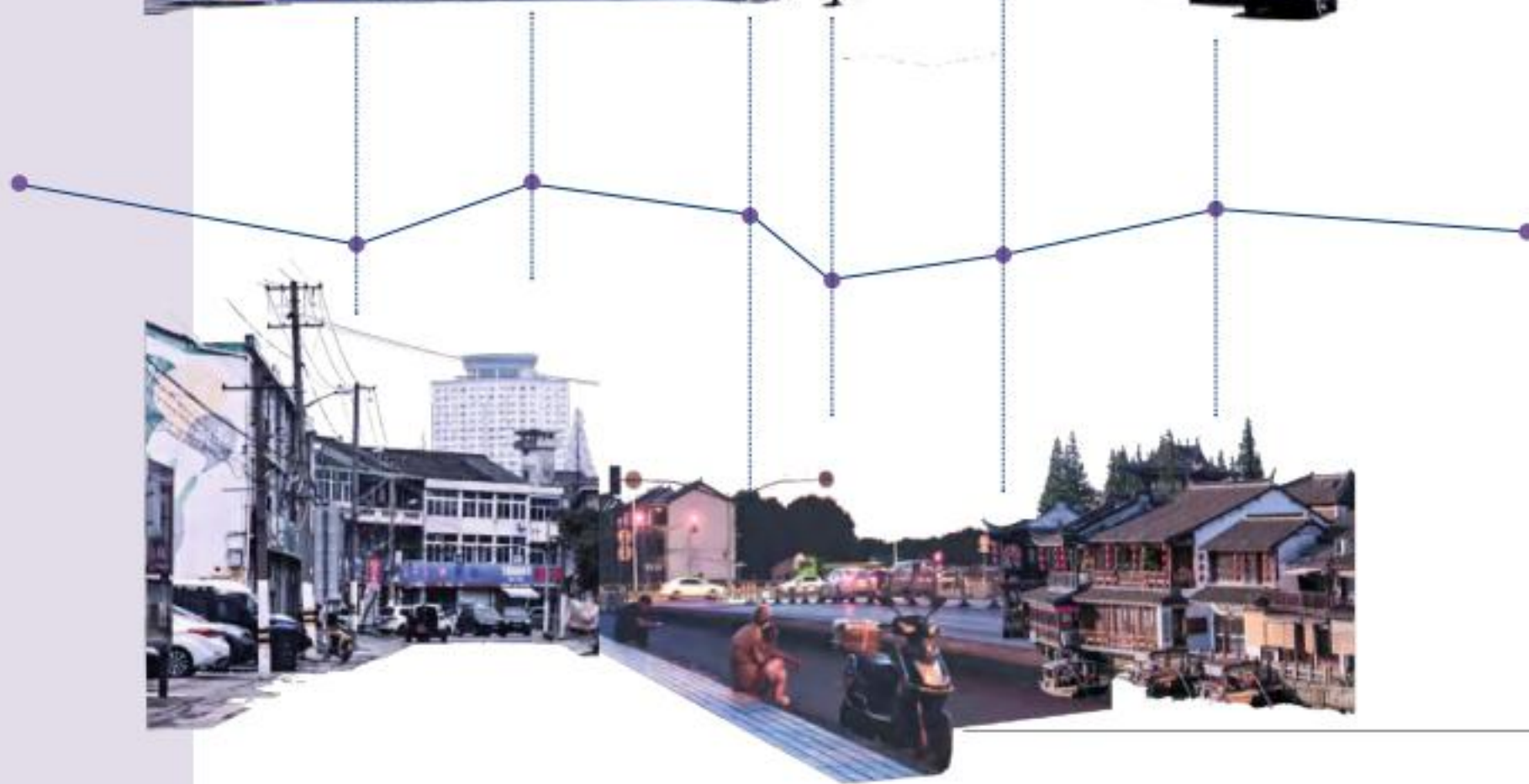
视线分析



轴测效果图



手工模型照片



02

城市节点振兴：基于 QGIS 的上海 地铁站聚类分析与评估

Urban Nodes Revitalization: Clustering Analysis and
Evaluation of Shanghai Metro Stations Based on QGIS

Urban Nodes Revitalization

Clustering Analysis and Evaluation of Shanghai Metro Stations Based on QGIS

城市节点振兴：基于 QGIS 的上海地铁站聚类分析与评估

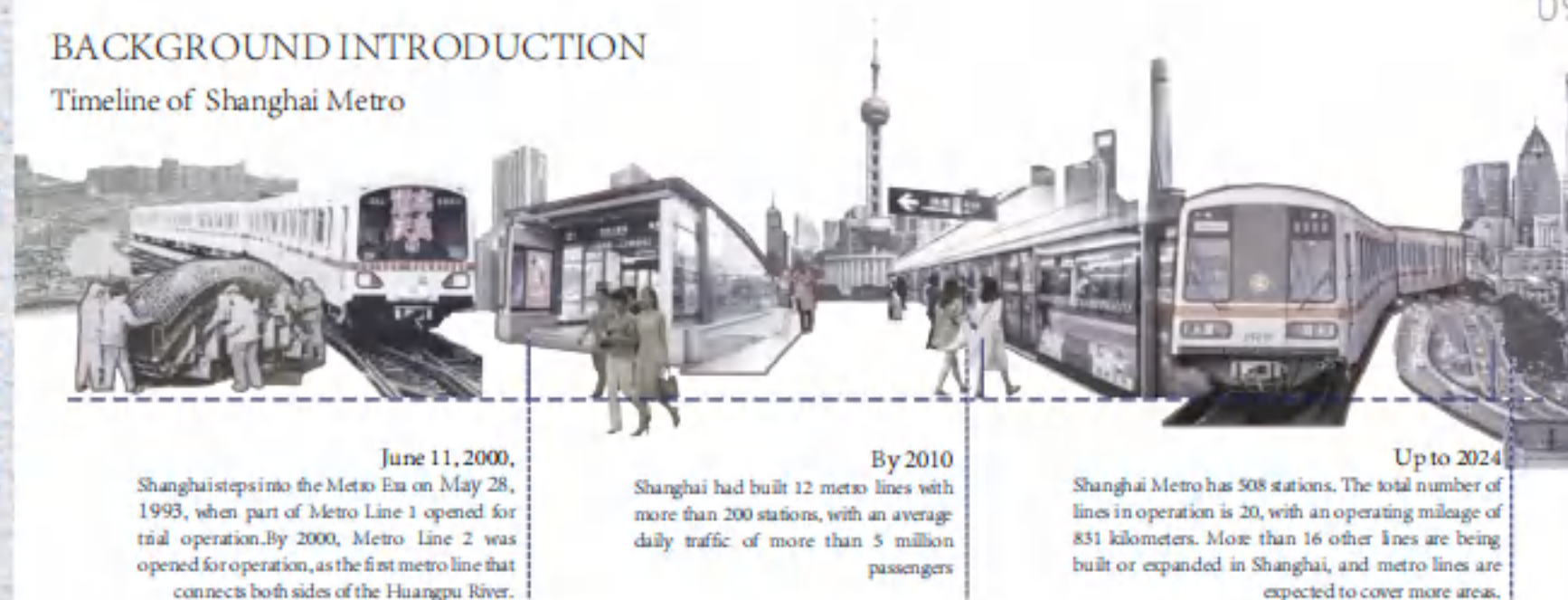
个人研究项目

本研究通过通过对上海地铁站周边的 POI 进行系统的 GIS 分析，探讨站点功能分布的现状并识别出多种具有代表性的站点类型，从而更深层次地认识城市，并为未来的城市规划和站点周边区域改造提供科学依据和针对性的规划提升建议。

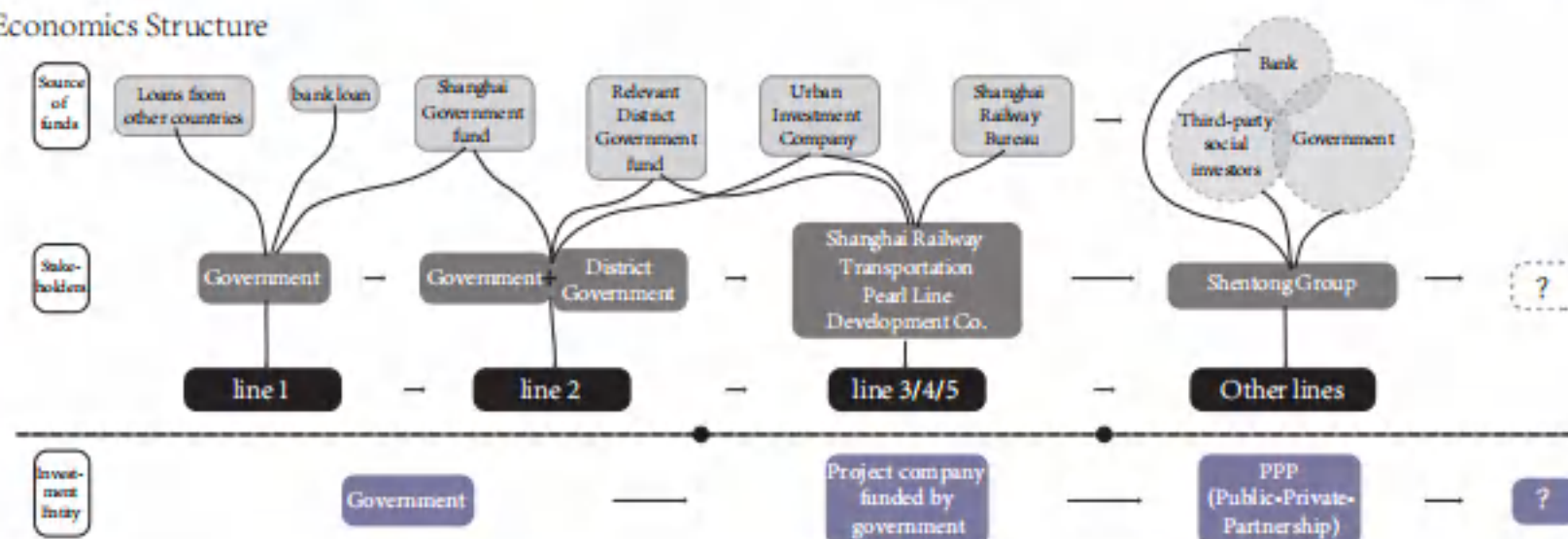


BACKGROUND INTRODUCTION

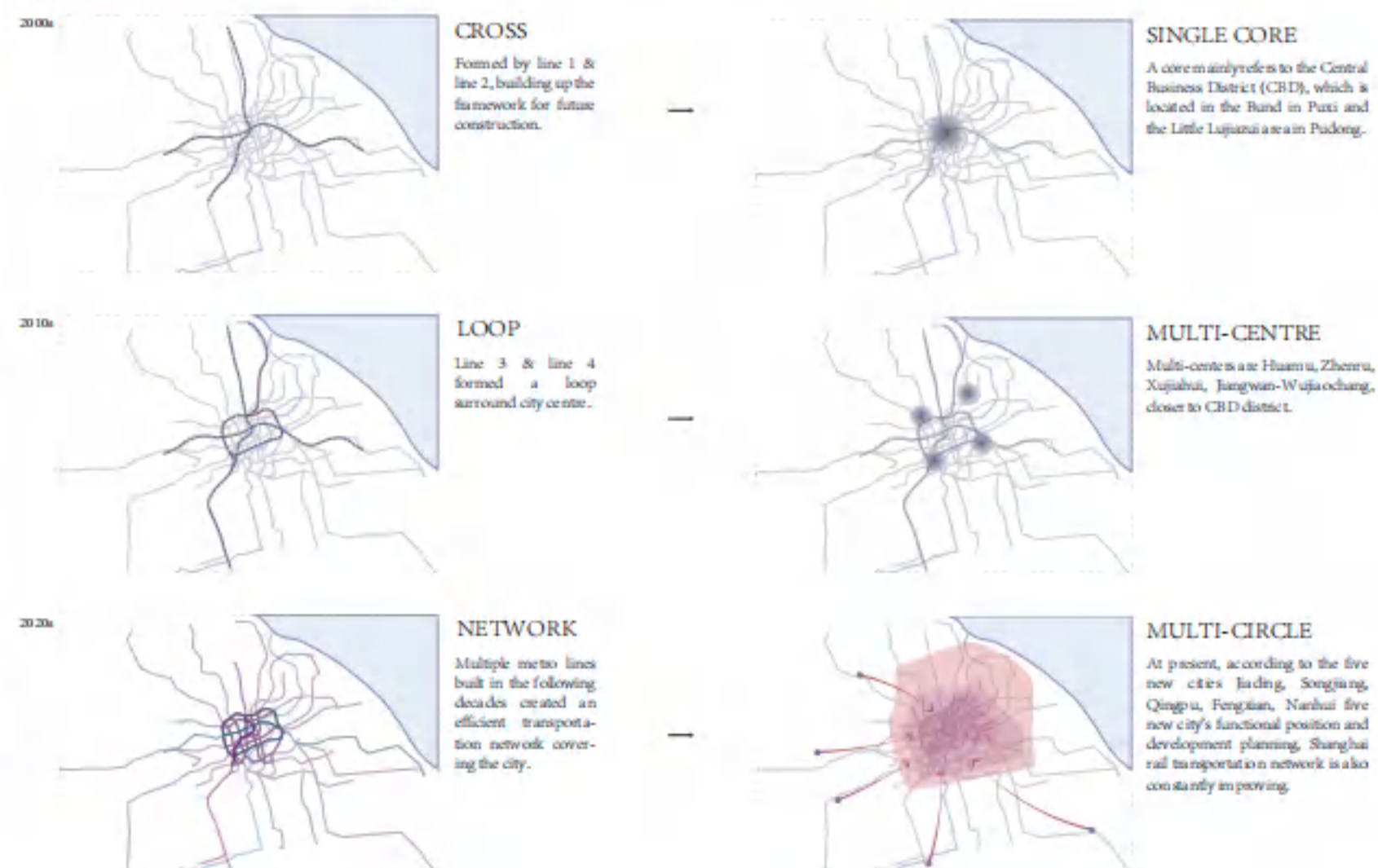
Timeline of Shanghai Metro



Economics Structure

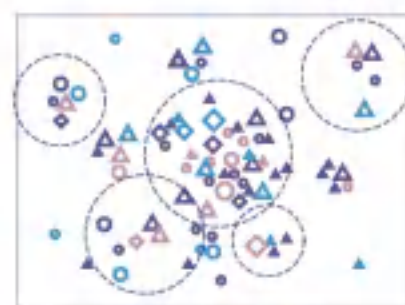


Metro&Urban Structure Evolvment



ANALYSIS PROCEDURE

The goal is to support transit-oriented development by understanding and addressing the unique needs of the environments surrounding metro stations, and provide targeted planning and redevelopment strategies for typical stations



Data analysis

Conduct a systematic analysis of decisive data that can assess the environment around the subway.



Typology categorization

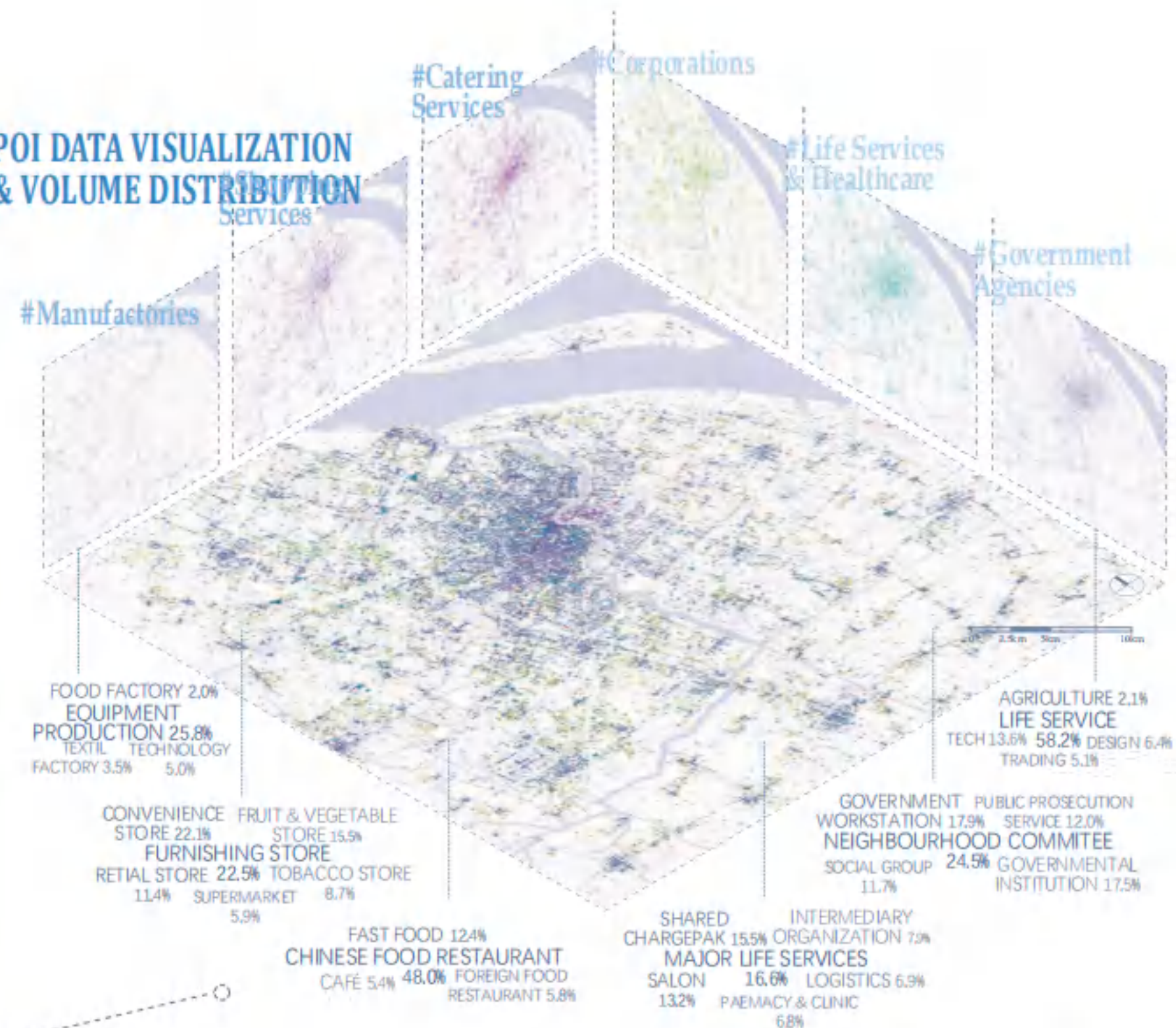
Explore the current functional distribution of these stations and identifying various representative station typologies.



Strategy outlook

Provide a scientific basis and practical guidance for future urban planning and to propose targeted planning and renovation strategies.

POI DATA VISUALIZATION & VOLUME DISTRIBUTION



STEP 1: DATA PREPARATION

a. Framework of physical factors.

Considering the spatial attributes that make up a well-functioning people-city, the initial physical factors to consider should be: Administration & Public Services, Administration & Public Services, Employment & Economic Activity.

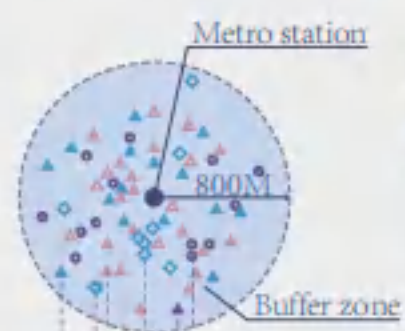
Evaluation Aspect	POI Categories in Gaode	Detailed Description	Data Interpretation
Administration & Public Services	Life Services & Healthcare	salon, intermediary organization, pharmacy&clinic, etc.	Reflect the quality of life and accessibility to healthcare services for residents.
	Government Agencies	neighbourhood committee, governmental institutions, social group, etc.	Reflect the administrative functions and level of public services available around metro stations.
Business Vitality	Catering Services	chinese food, fast food, foreign food, coffee shop, etc.	The density of dining establishments is typically associated with pedestrian traffic, commercial value.
	Shopping Services	furnishing store, convenience stores, retail stores, supermarkets, etc.	Indicate the level of commercial services and the extent to which daily shopping needs are met.
Employment & Economic Activity	Corporations	life service, technology, design, agriculture, trading, etc.	Representing the concentration of commercial activities& employment opportunities.
	Manufactories	equipment production, textile factory, food factory, etc.	Reveal the industrial character and employment opportunities near metro stations.

b. Visualization and election of appropriate POI types that reflect the functional richness and convenience of businesses around subway stations.

STEP 2: CREATING BUFFER

a. Creating Buffer-zone:

Based on the 15-Min Pedestrian Neighborhood theory, a buffer zone is established with a radius of 800 meters.



b. Calculating numbers of POIs in each buffer.
eg. POI number in East Nanjing road buffer.

STATION	Life	MANU	SHOP	CATER	CORP	Life	GOVERN
East Nanjing road	2-10	1	962	1538	520	132	219

STEP 3: CLUSTERING ANALYSIS

a. Feature Normalization.

Use Standardization Formula:

$$Y = (X - \mu) / \sigma \sim N(0,1)$$

to transform the distribution of elements along the variables so that different kinds of data become comparable.

Clustering algorithms: K-means

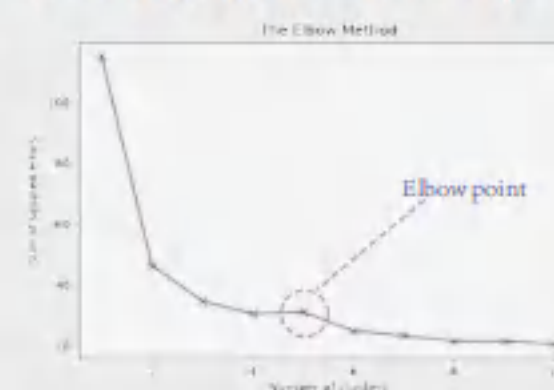
Number of cluster: 5

Normalize attributes: Yes

Number of times to run K-Means: 50

b. Implementation of clustering algorithm.

Put all POI data into **K-means** clustering algorithm to classify.



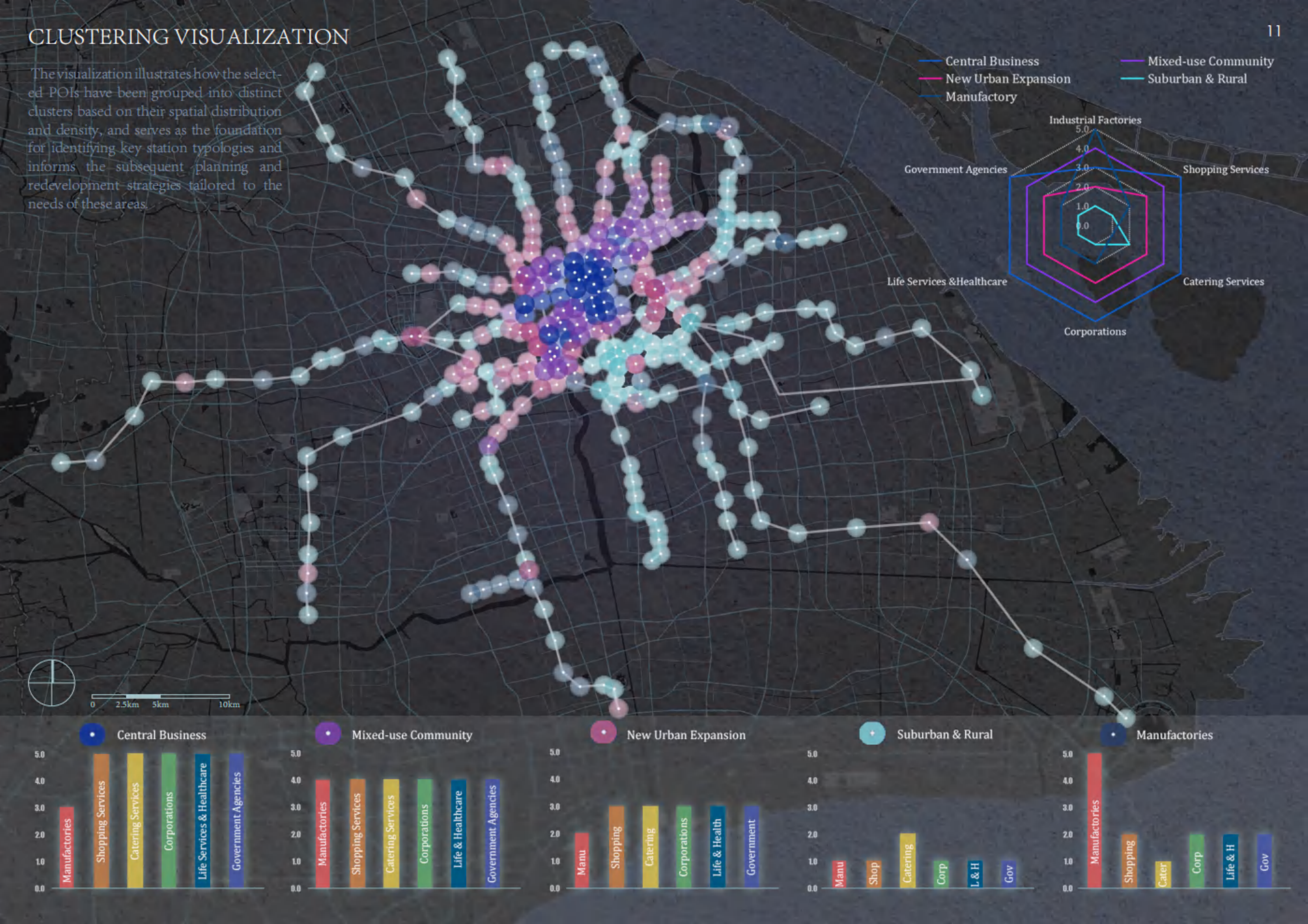
c. Centre of clusters:

This table shows the average number of poi in the buffer zone of five types of subway stations. Then assign values by sort in order from 1 to 5 to show the attributes of each cluster.

class	Admin	Industrial	Shopping	Catering	Corporations	Life Services & Healthcare	Government Agencies
A	35	1.2	419.4	793.5	307.7	936.2	210.4
B	55	1.7	244.3	391.2	165.8	535.2	157.6
C	77	0.6	141.4	239.2	77.1	290.2	73.8
D	146	0.2	42.2	57.4	22.9	76.7	23.0
E	49	3.2	56.9	54.0	52.8	82.9	24.4

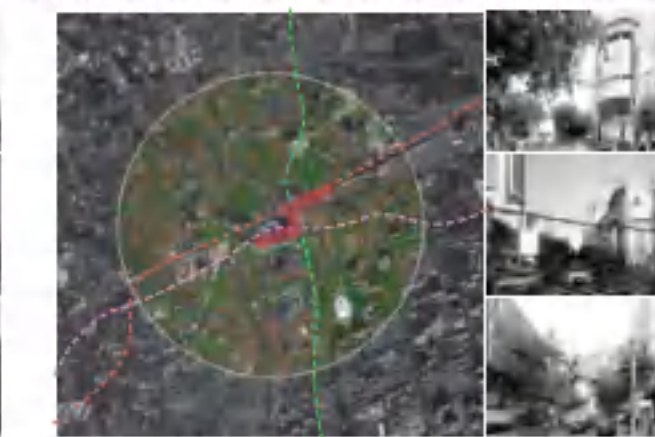
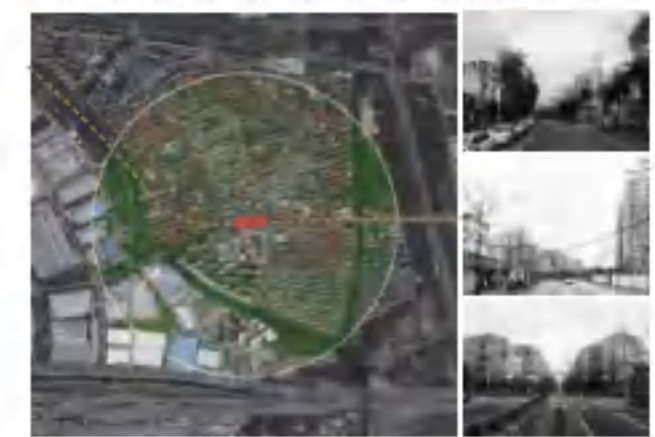
CLUSTERING VISUALIZATION

The visualization illustrates how the selected POIs have been grouped into distinct clusters based on their spatial distribution and density, and serves as the foundation for identifying key station typologies and informs the subsequent planning and redevelopment strategies tailored to the needs of these areas.



TYOLOGY ANALYSIS

An in-depth exploration of the spatial and functional characteristics of various metro station typologies in Shanghai, aimed at identifying key patterns and informing future urban planning strategies.

TYPOLOGY TYPE					
DATA FEATURE	Central Business a. Service levels in these two types are among the highest across all five categories. b. Many adopted the Transit-Oriented Development (TOD) approach, directly linking metro projects with real estate developments such as shopping malls.	Mixed-use Community	New Urban Expansion a. Service levels are at a mid-range level. b. Have significant potential for improvement in terms of community amenities, commercial services, and transportation accessibility.	Suburban & Rural a. Service levels are relatively low, with dense populations and poor-quality service facilities. b. Significant jobs-housing imbalance, lack of employment opportunities, causing high commuting volumes.	Manufacture a. Strong industrial presence. b. Quality of service facilities is relatively low to mid-range. c. Noticeable jobs-housing imbalance, heavy reliance on metro stations.
CASE STUDY & STRATEGY	 a. Service levels in these two types are among the highest across all five categories. b. Many adopted the Transit-Oriented Development (TOD) approach, directly linking metro projects with real estate developments such as shopping malls. Overall, the POI indicators for these station types align well with their functional roles, indicating no urgent need for renovation or redevelopment. Therefore, no specific improvement strategies are proposed for these station types.		 a. Service levels are at a mid-range level. b. Have significant potential for improvement in terms of community amenities, commercial services, and transportation accessibility. a. Improve transportation accessibility, create new TOD centers as core connections. b. Enhance its overall life quality and attractiveness.	 a. Service levels are relatively low, with dense populations and poor-quality service facilities. b. Significant jobs-housing imbalance, causing high commuting volumes. a. Introducing commercial and employment opportunities, reduce commuting demands. b. Increase regional self-sufficiency.	 a. Strong industrial presence. b. Quality of service facilities is relatively low to mid-range. c. Noticeable jobs-housing imbalance, heavy reliance on metro stations. a. Upgrade the quality of living services for a better jobs-housing balance. b. Enhance surrounding environment quality.



03

URBAN ESCAPE

_FUXING ISLAND URBAN DESIGN

2024 CAUP International Summer School
Design Work

同济大学 2024 暑期国际夏令营项目

指导老师: Prof. 黄璜, Prof. Emilie Gascon, Prof. Jose
Manuel Almondovar Melendo.



- Fuxing Island
- Huangpu River
- Inner ring
- Middle ring
- Outer ring
- Airport

URBAN ESCAPE

— FUXING ISLAND URBAN DESIGN

2024 CAUP International Summer School Design Work

同济大学 2024 暑期国际夏令营项目

指导老师: Prof. 黄璜, Prof. Emilie Gascon, Prof. Jose Manuel Almondovar Melendo.

本项目旨在将位于上海杨浦区的复兴岛转变为一个生态友好的休闲地，设计充分利用岛上的水体、工业遗产和生态潜力，营造出与自然和谐共存的空间。

设计愿景：主题“URBAN ESCAPE”旨在为游客和居民提供慢节奏生活的机会，通过多样化的出行方式、休闲设施以及放松空间，促进身心健康。策略结合生态保护、工业历史的融合与健康生活的推广，创造出一个既能让人逃离都市喧嚣、又保持岛屿独特身份的城市绿洲。

This project aims to provide both visitors and residents with an opportunity to experience a slower pace of life, promoting well-being through diverse mobility options, recreational facilities, and spaces for relaxation. The strategy combines ecological preservation, industrial history integration, and the promotion of healthy living, offering an urban escape while maintaining the island's identity.



- Fuxing Island
- Gongqing Road (access to island)
- Middle Ring (Shanghai)
- Local ferry
- High priority road
- Huanpu River
- Metro line 12

Mobility: connected, but disconnected



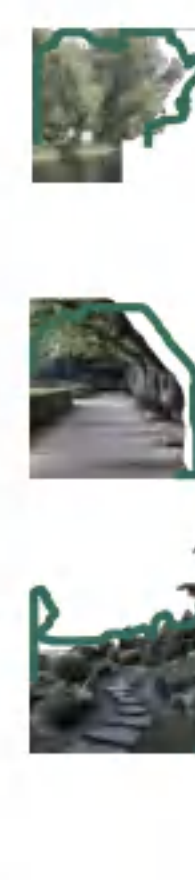
- Technology and Service
- Residential
- Industry

Functions: characterized by industry



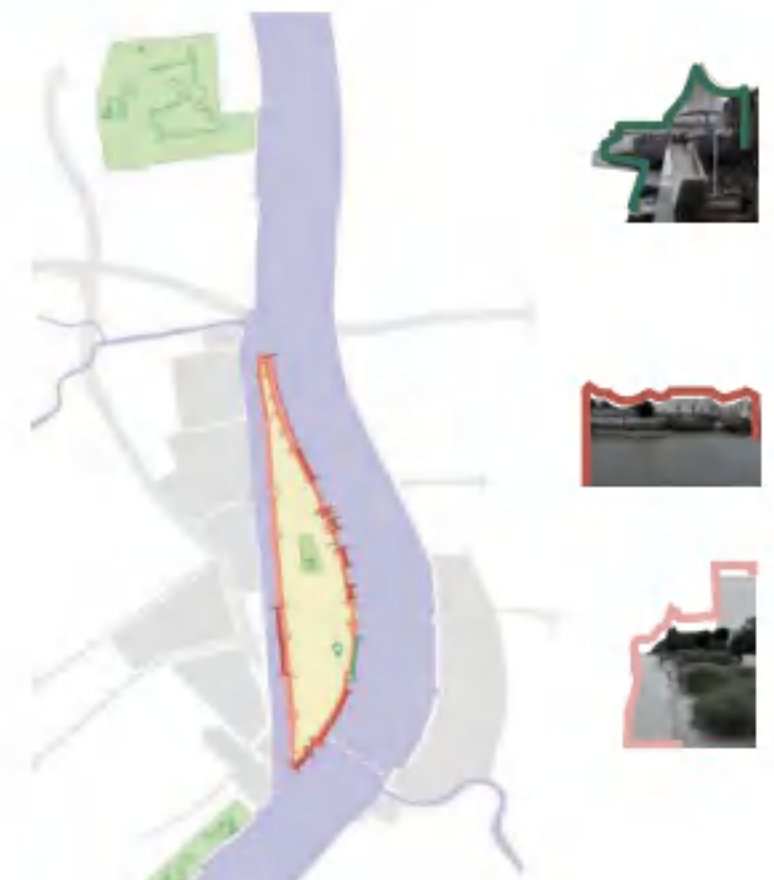
- Local recreation

Green: patches, but no connection



- Waterfront sidewalk
- Waterfront wetland
- Waterfront blocked by wall
- Port areas

Waterfront: heavy industry, little access



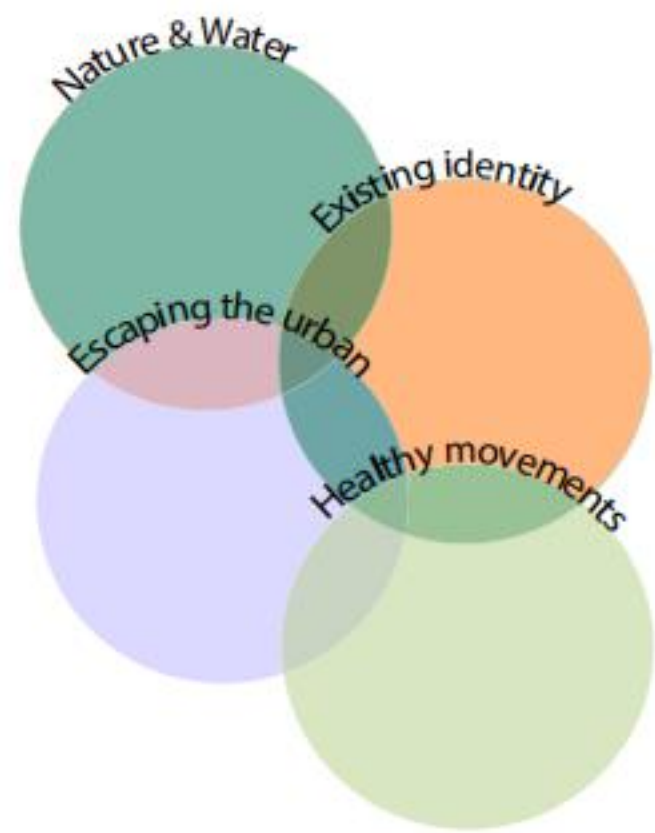
Derived potential of Fuxing Island



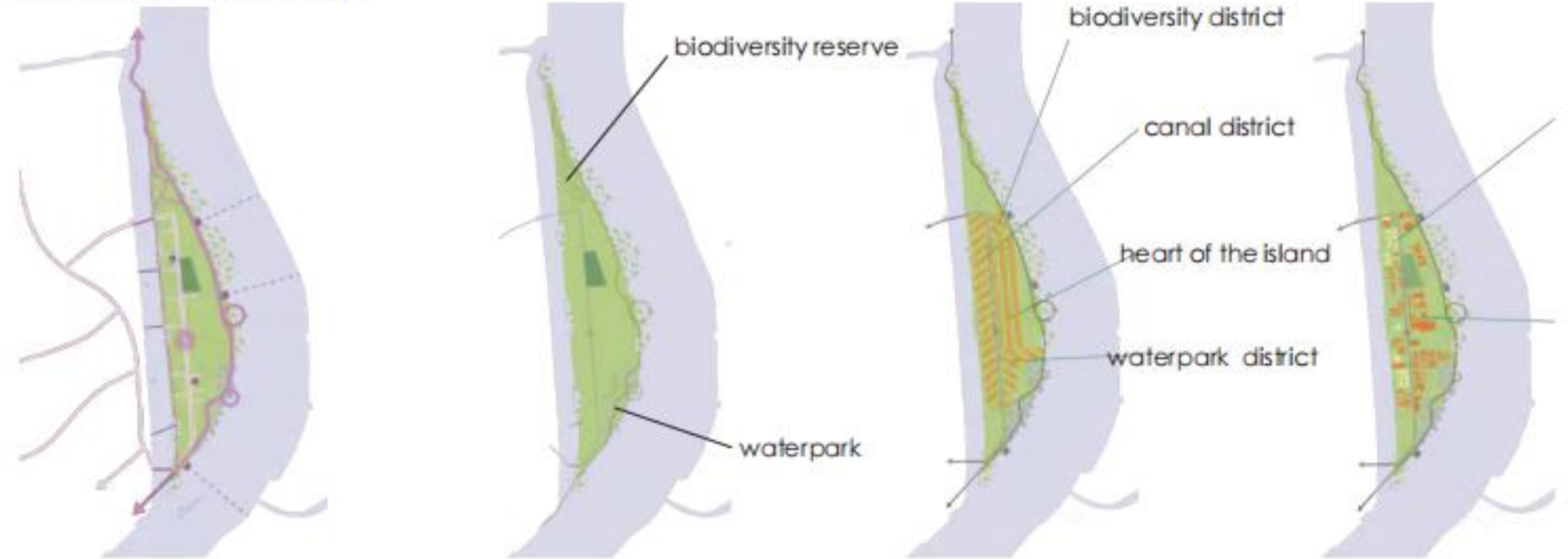
- Natural point of interest
- Economic and educational point of interest
- Residential/neighboring districts

VISION DESCRIPTION

- _ With, not against water & nature [Resilience & ecology]
 - _ Healthy movements [Accessibility & mobility]
 - _ Working with the existing [Connecting industrial history, buildings, nature & water]
- Escaping from the generic urban into a new healthy urban



Design Strategy



- Accessibility & healthy movement**
 - Bridge
 - Main street
 - Street
 - Main bike- & footpath
 - Metro Station
 - Parking
- Resilience & ecology**
 - Fuxing Island Park
 - Districts**
 - Main bike & footpath
 - Water retention ponds
 - Water treatment wetlands
- Central district**
- Working with the existing**
 - Ferry station
 - Metro station
 - New buildings
 - Existing buildings

VISION: A PLACE FOR ALL





MASTERPLAN: FUXING ISLAND

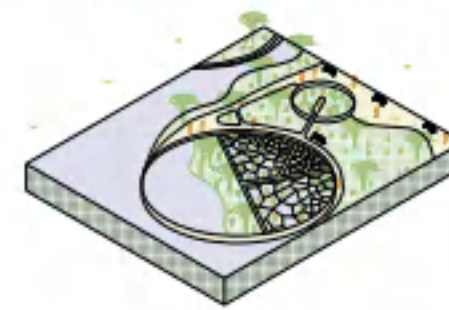


Natural bridges: new pedestrian walks

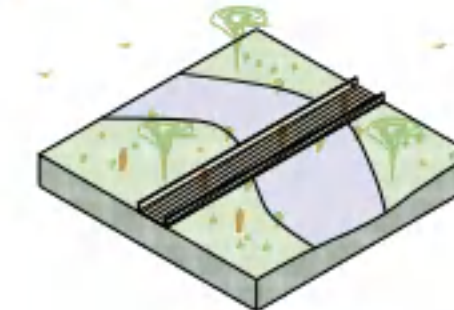


Water Tank: Ecological education and community engagement

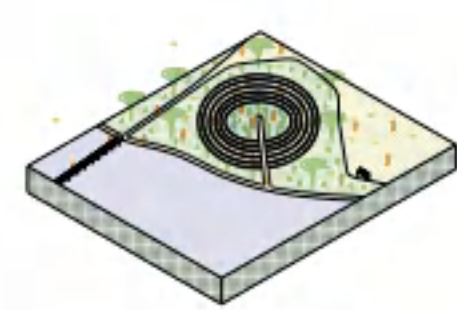
ACTIVE BLUE/GREEN TOOLBOX:



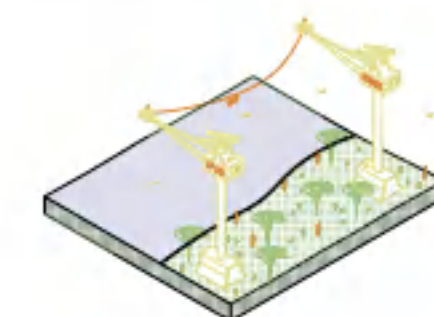
Water platform: enjoy the biodiversity



Natural bridges: new pedestrian walks



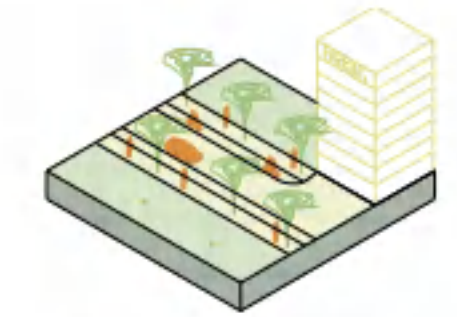
Green plaza: changing functions



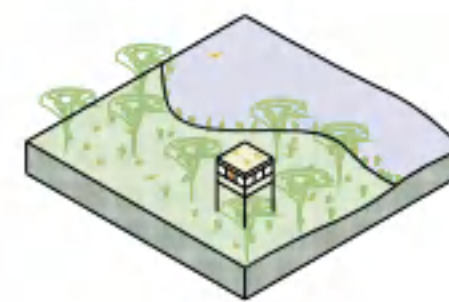
Reused heritage: zip lines & restaurants



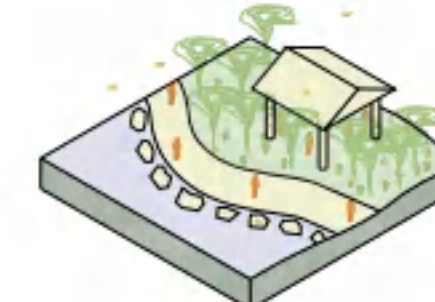
Mixed use bridge: enjoy the mobility



Street sharing: cars go in garages



Wilderness: let non-humans be free



Chinese garden: enjoy nature's peace



New Connections: Walk without fear



04

城市建筑设计 II：居住类建筑设计 Urban Architecture II: Residential Architecture Design

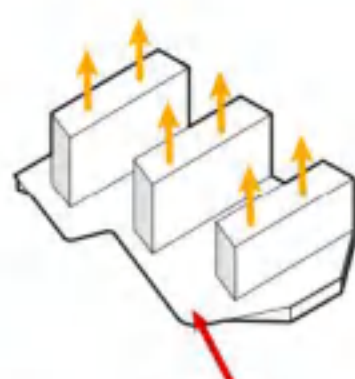
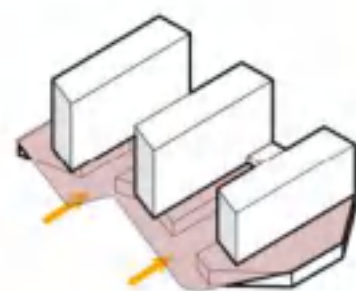
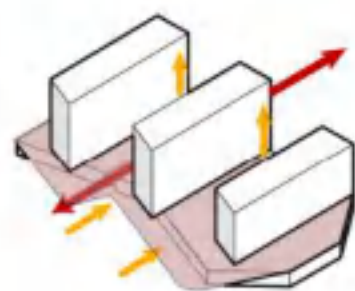
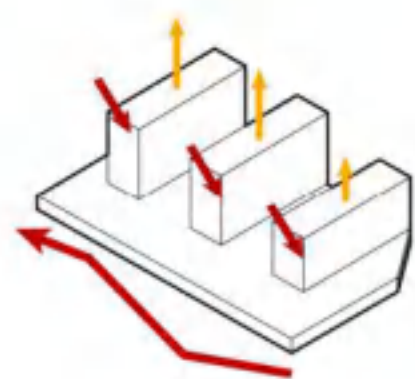
城市设计专业大二下课程设计

指导老师：贺 永 王志军 张灵珠

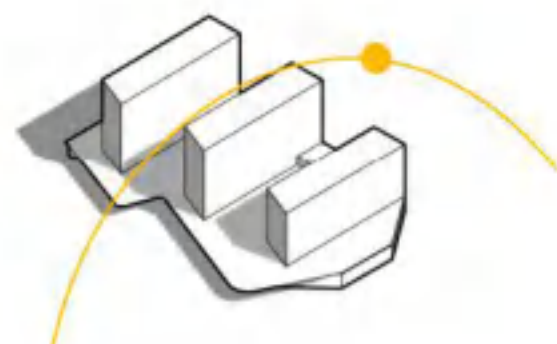
小组成员：徐晓驰 刘瑾一 谭皓纯



轴测效果图



垂直交通



光照情况

体块生成

城市建筑设计 II

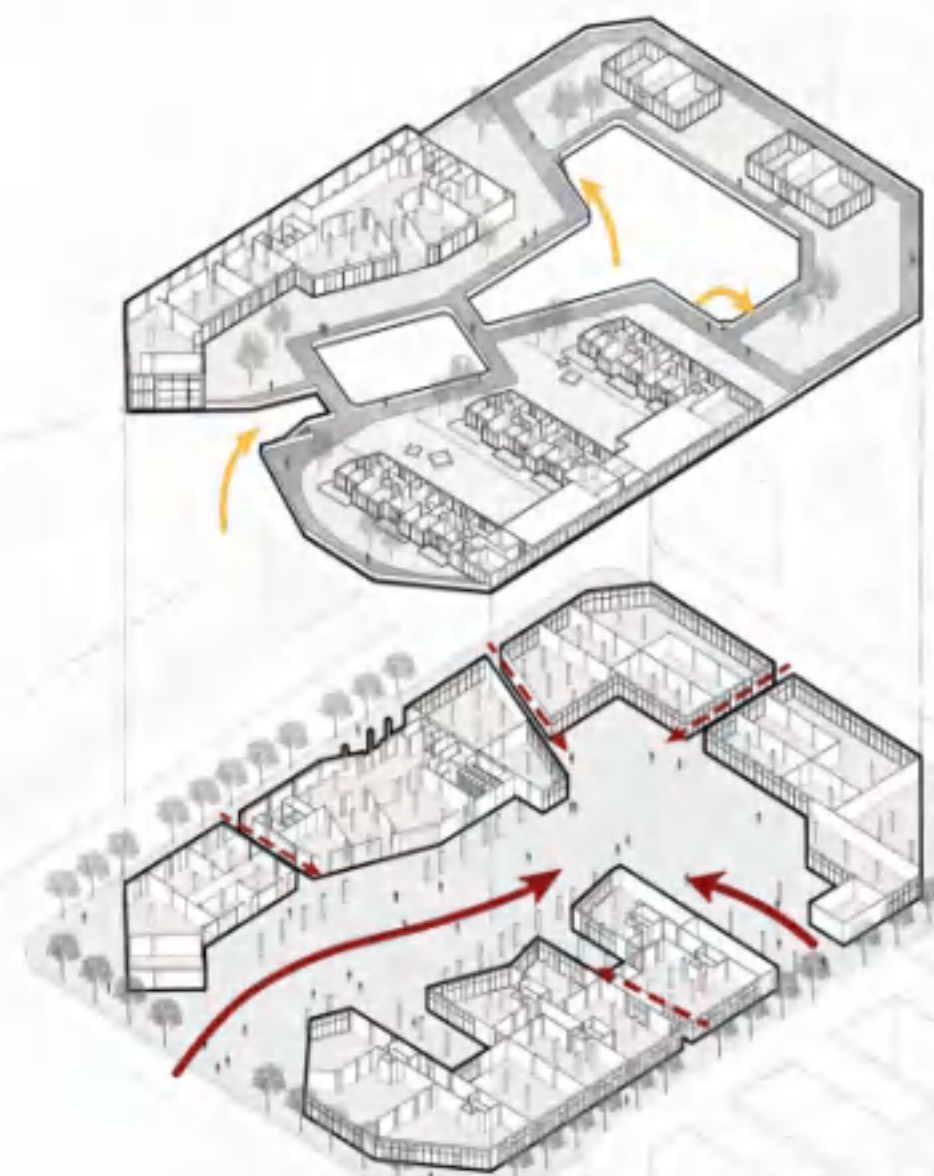
——居住类建筑设计

Residential Architecture Design

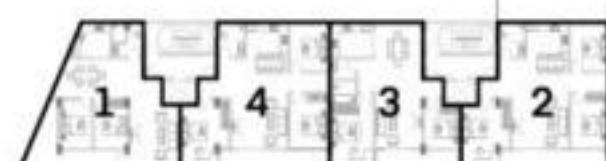
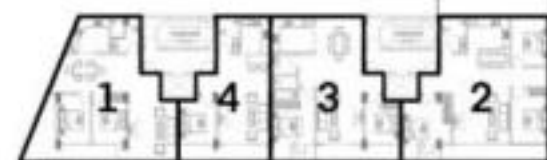
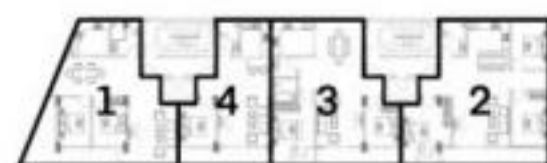
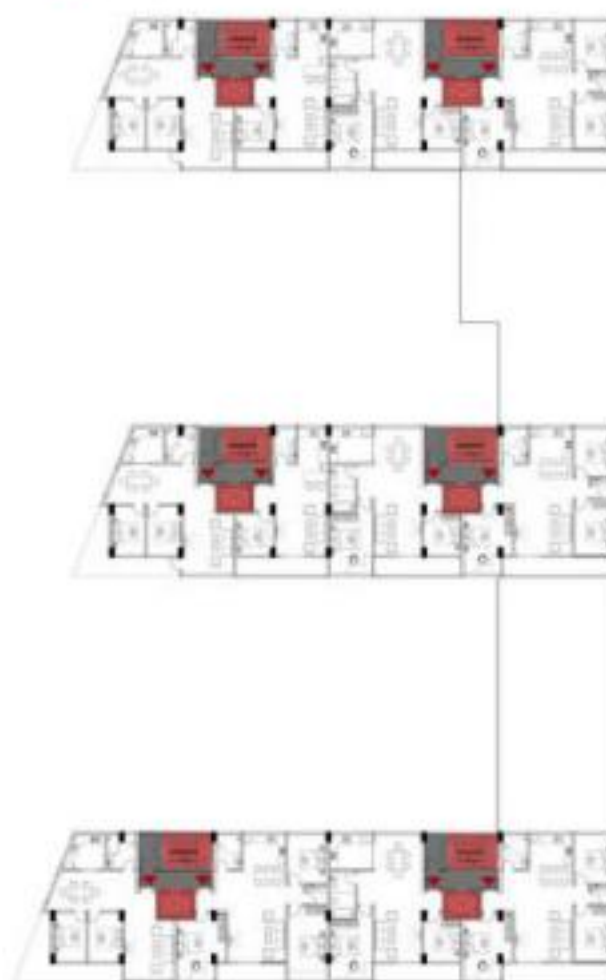
城市设计专业大二下课程设计

设计目的：在于了解集合住宅的类型和特点、相关建筑的功能组成、空间组织特点、建筑容量与用地尺度，并展开对城市未来住宅形式的畅想。

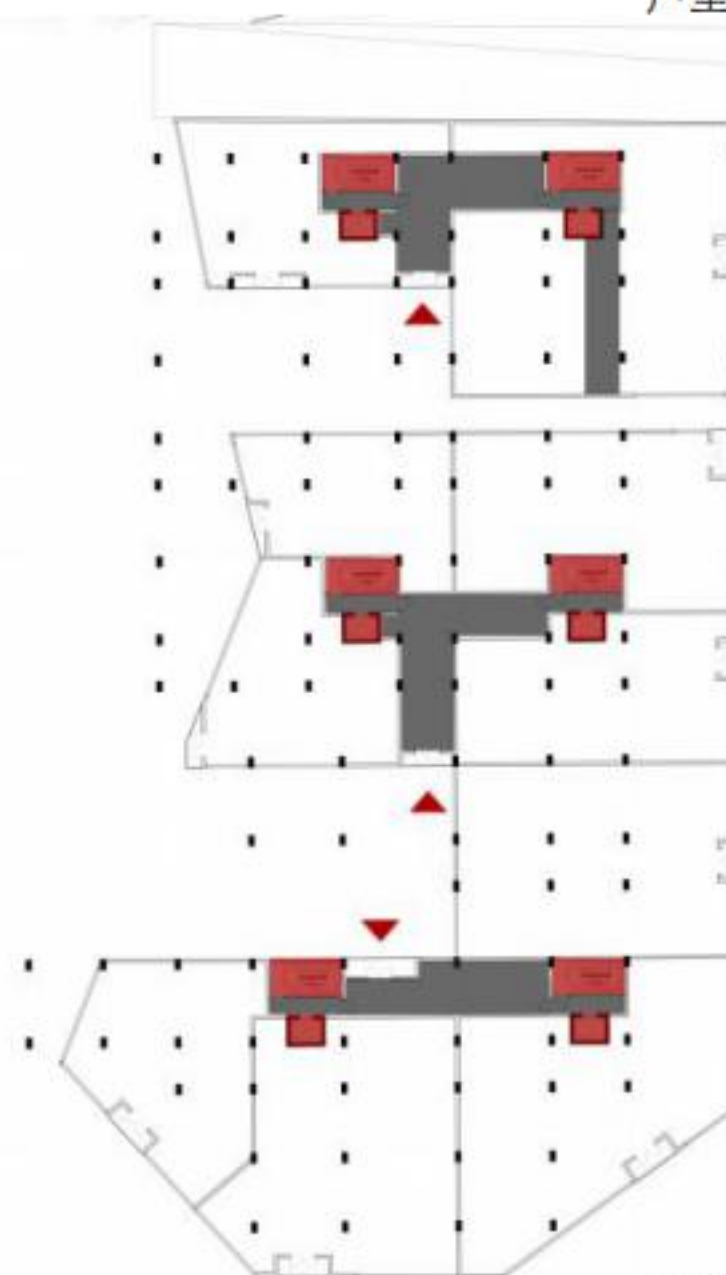
设计说明：设计意图在于打破居住社区与城市的边界，使居住社区与城市在空间与交往层面相融合。小组部分形态为三个地块向内围合，功能上底层为商业，二层以上为住宅，垂直方向上设置两层交通平台，商户和城市居民可以在一层进行商业活动和聚集，住户则可以从二层平台或内部电梯进入居住区域。个人部分为板式住宅设计，设置四种标准户型分别对应不同家庭需求，均为南北通透、一梯两户、开放阳台设计。



分解轴测图



户型分析



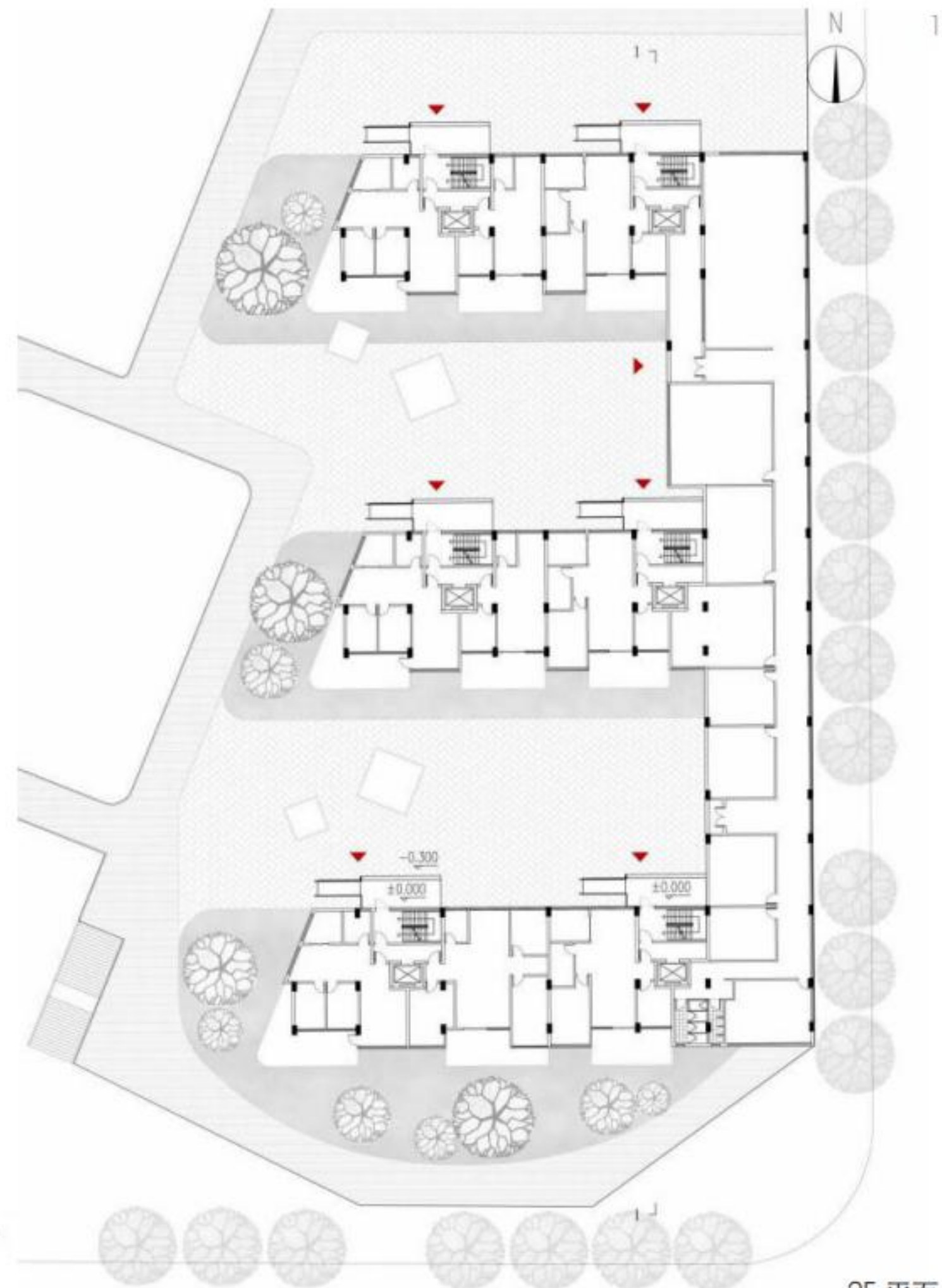
流线分析

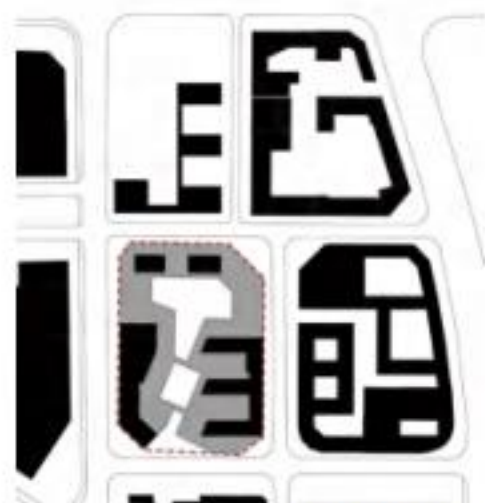


电梯



户型平面

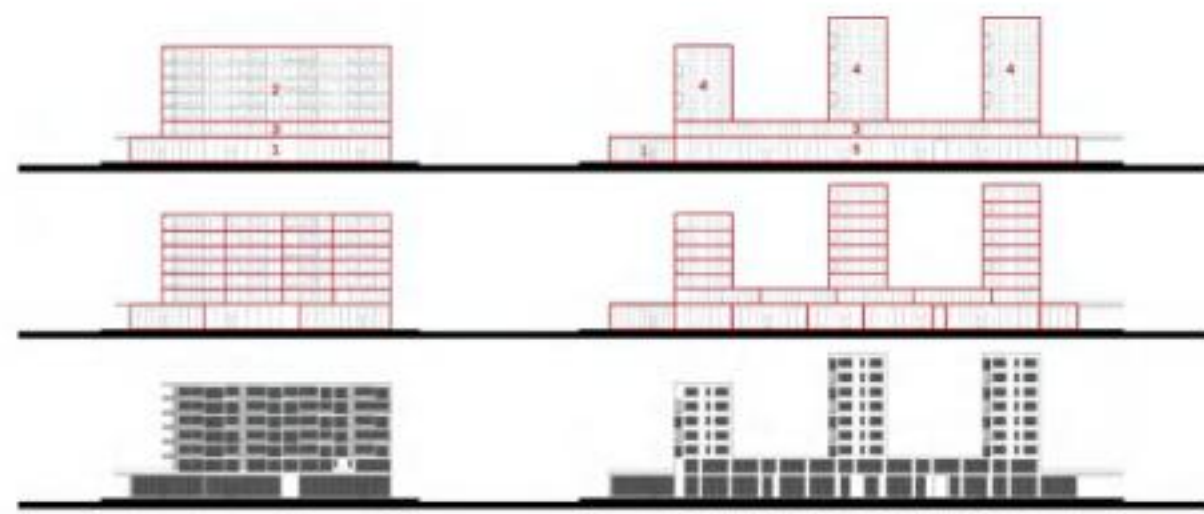




区位示意



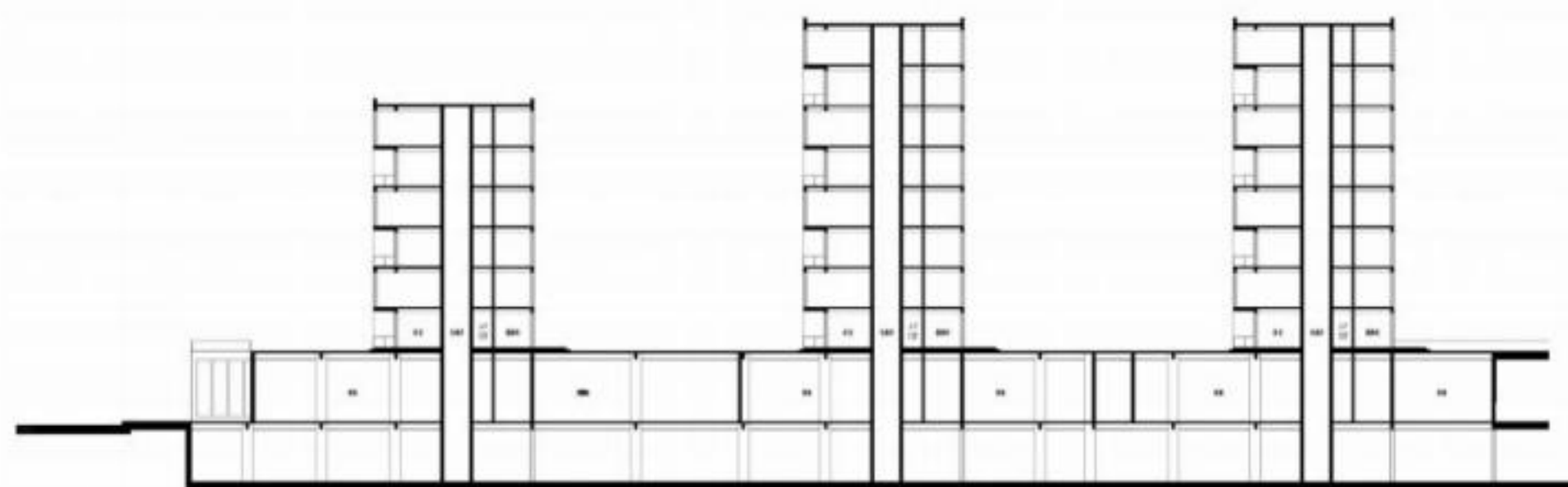
光照分析



立面分析



南立面 1: 1000



车行道 人行道

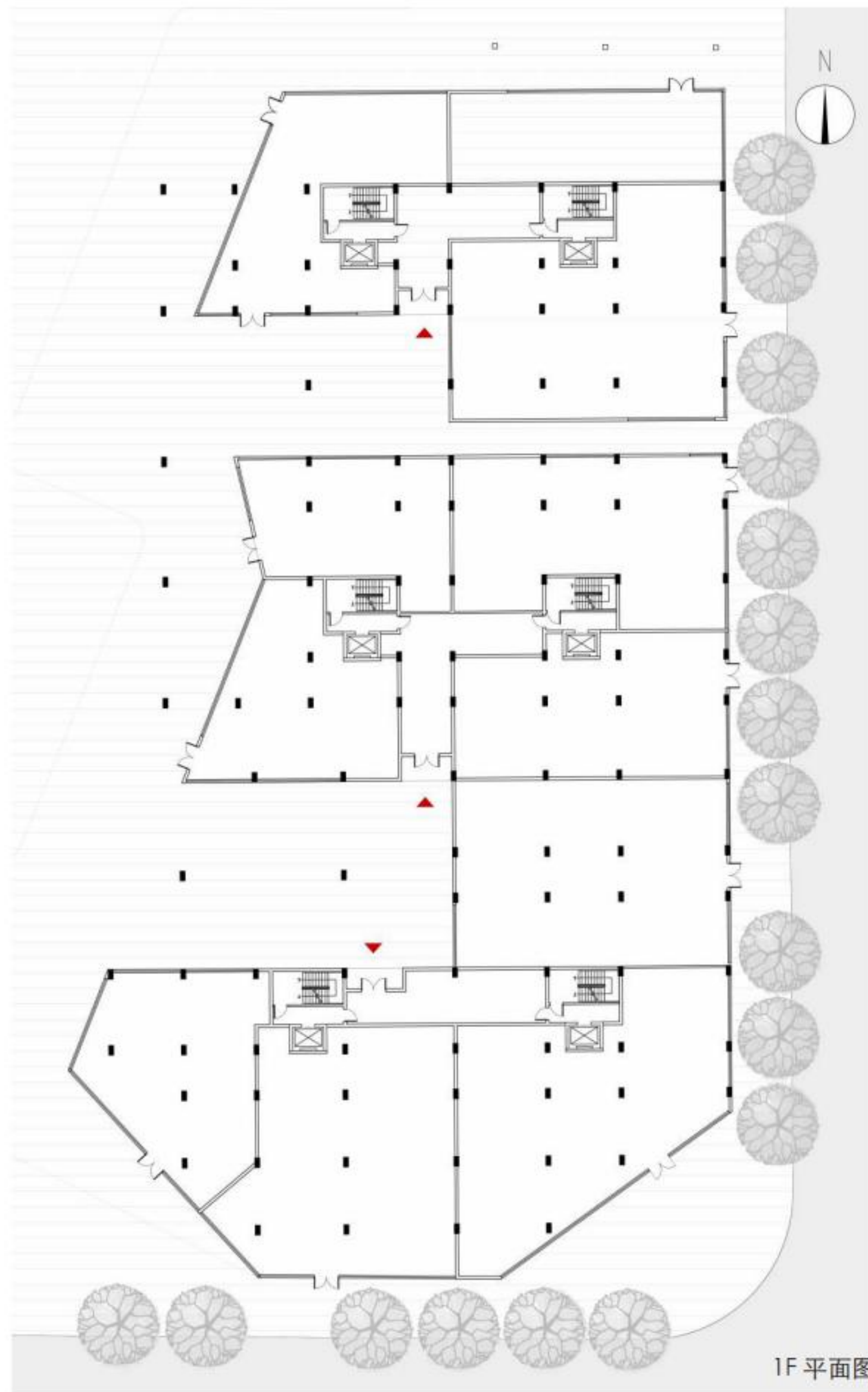
建筑主体 (含二层以上公共平台)

内部广场

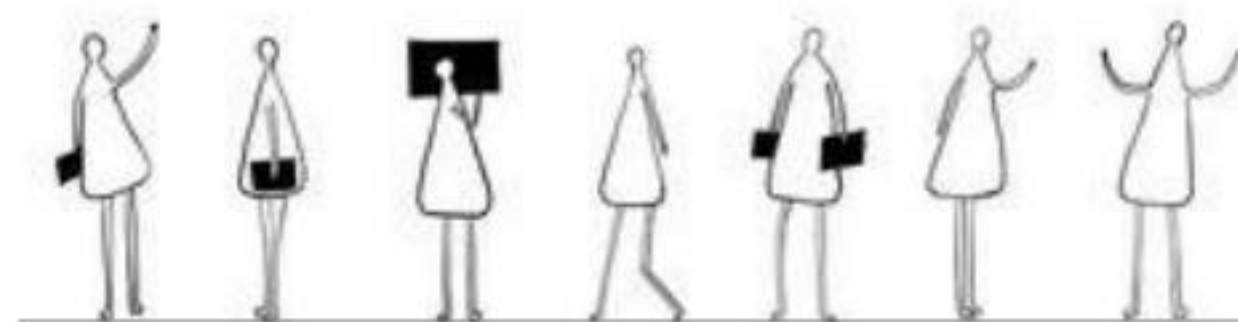
街块剖面 1: 1000



东立面 1: 1000



1F 平面图



05
other works

城市建筑设计 II

——社区活动中心建筑设计

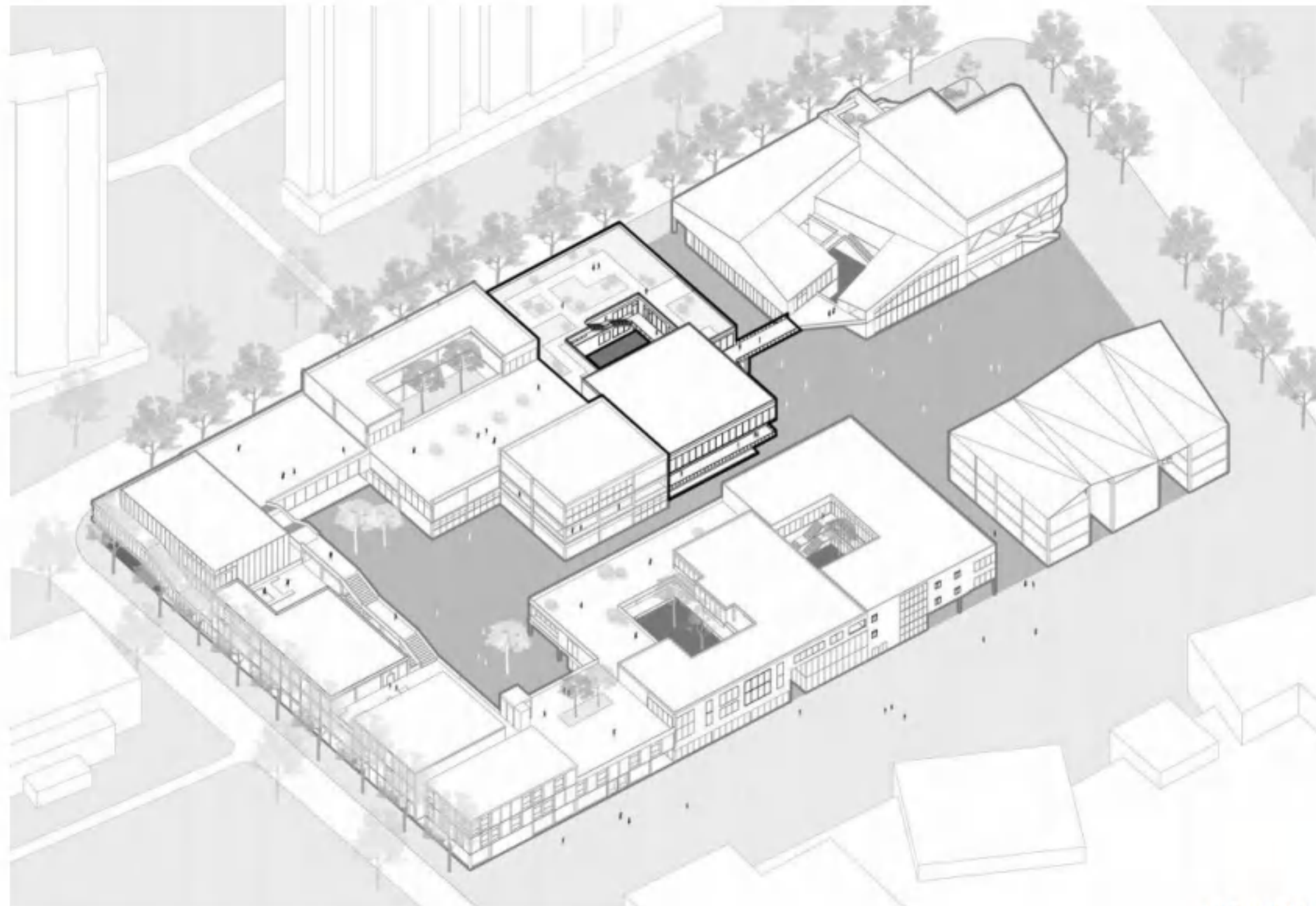
Community Center Architecture Design

城市设计专业大二下课程设计

指导老师: 王志军 贺 永 张灵珠

设计目的: 通过社区服务中心各类不同功能的小型公共建筑设计, 掌握公共建筑设计基本原理, 并了解相关建筑的功能组成、流线组织、空间特点、建筑容量与用地尺度的基本关系。

设计思路: 小共同确定好公共空间体系及广场内部用地红线, 设置内外广场 + 二层三层连廊的公共空间布局, 形成街道和广场, 并与保留的现有建筑形成统一的社区公共活动空间体系。个人设计部分, 主要以服务类功能为主, 内部设置次级公共空间, 并通过连廊和三层屋顶花园与整体的公共空间体系相连。



小组轴测图



1F 平面图



2F 平面图



3F 平面图



立面图



剖面图

大三上课程设计：国康路街道设计 Guokang Road Street Design

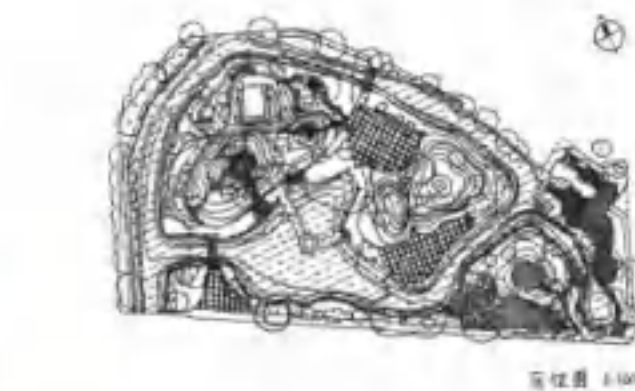
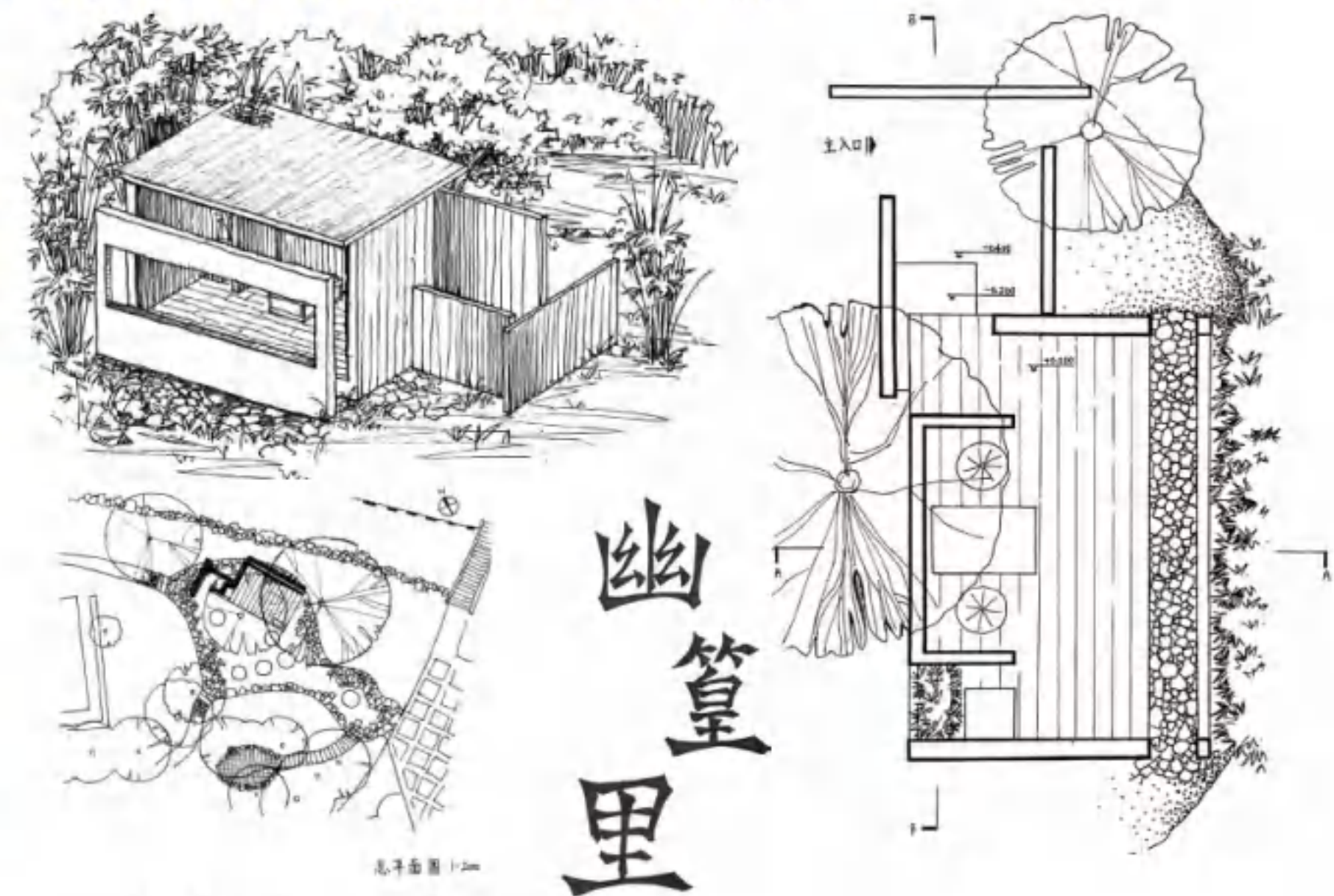
Analysis & Strategy



Individual Design

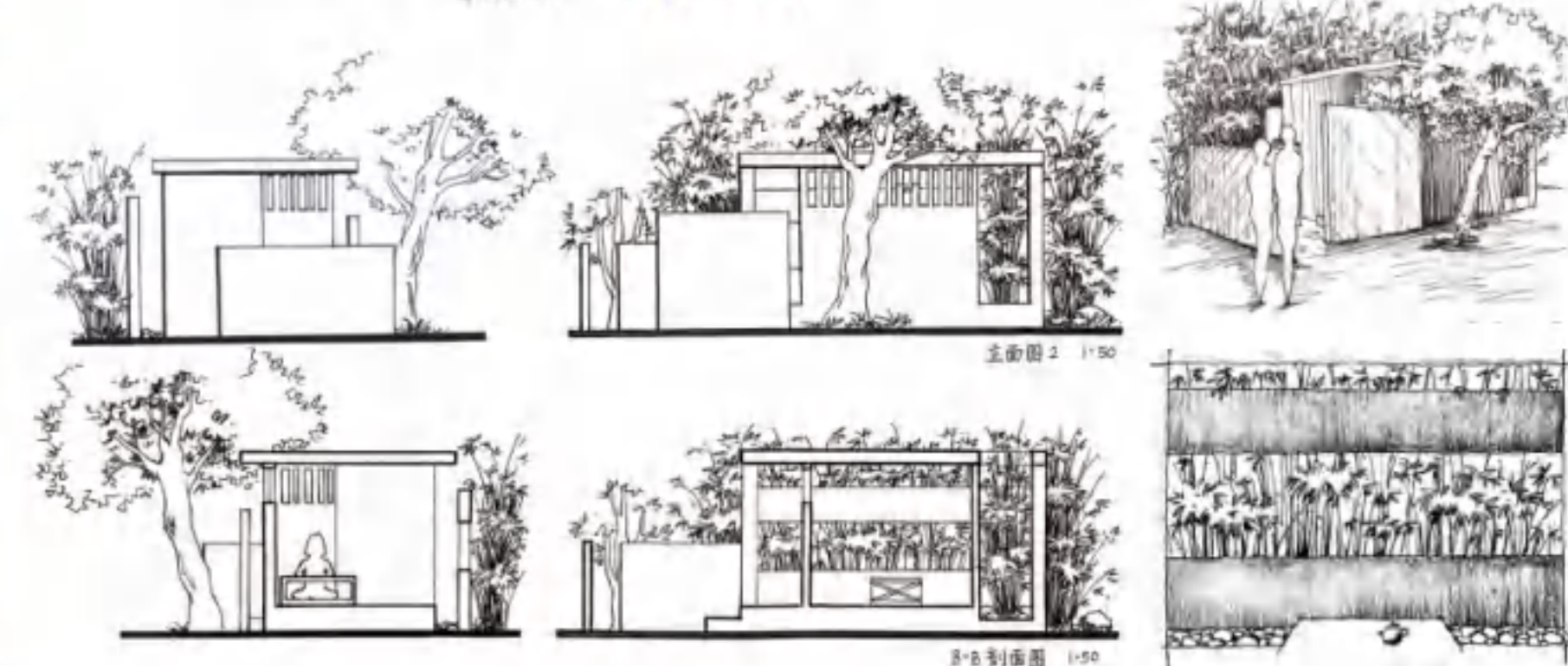


20 级转专业暑期课程包：三好坞茶室设计 Teahouse Design in Sanhaowu



设计说明：

丛丛竹林掩映藏于深处的一方茶室，寂静幽秘。茶室入口处位于竹林最深处，主体空间为敞开式，东南侧独立于主体空间单独设立一面墙，墙上的横向长窗成为了墙后竹林的取景框。茶室内可容二人席地坐于竹垫之上，有竹林环绕，“独坐幽篁里”之感，相谈品茗，静听竹吟风啸。



模型制作 Modeling



绘画作品 Drawings



写生速写、水彩



工笔国画



速写



素描



Xiaochi Xu

• Education

Tongji University, Shanghai

Bachelor of Engineering, Urban Design

GPA: 4.68/5 91.83/100

2020.9-2025.7

• Academic Experience

2nd Level Scholarship, Tongji University

2021-22/2022-23

International Design Summer Camp, Vienna, Prague, Berlin

2023.8-10

TU Wien, CTU | Supervisor: Prof. Kai Xu, Prof. Gang Liu

- Designed for Arsenal New City Transformation.
- Researched in Vienna, Prague, Berlin Urban History and Morphology.

CAUP International Summer School, Shanghai

2021.5-6

Tongji University | Supervisor: Prof. Emilie Gascon, Prof. Jose Manuel Almondovar Melendo, Prof. Huang Huang.

- Developed innovative urban planning solutions that promote health and resilience in Cities, focusing on creating comprehensive designs for Yangpu Fuxing Island.
- As group leader, cooperated with students from 6 different countries, responsible for organising, idea gathering, design sketch, master plan drawing and presentation.

• Intern Experience

Urban Designer Intern: Jinqiao Real Estate Group

2023.7-8

- Participate in program discussion, project symposium and on-site research and investigation, eg. curtain wall panel Acceptance in Jinwan Wuqi Mountain project.
- Data collecting, text organizing and public tweets writing.

Freelance Intern: Landscape Space Sub-laboratory, Tongji University

2023.4-5

- Landscape Analysis and Renovation of Changwu Campus
- Site research, mapping and conceptual design; Responsible for master plan sketching using Illustrator.

• Social Experience

Outstanding Officer of Student Union

2022-2023

Student Development Department, CAUP Student Union: Head of Research and Study Planning Team.

- Organized and planned Office Tour and Citywalk.

Vice Monitor of Urban Design Class of 2020

2021-2023

- Assisting the class monitor in the annual evaluation of excellence and other class activities.

Volunteer: 2021 International Student Conference on Environment and Sustainability, UN

2021.6

